European Polymer Journal 192 (2023) 112068

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Recent advances in biodegradable polymers – Properties, applications and future prospects

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ARTICLE INFO

Keywords:
Biodegradable polymers
Classification
Fabrication
Properties
Applications

ABSTRACT

Polymers with biodegradable properties are termed as the future "green materials" possessing a shorter lifetime compared to synthetic polymeric materials. Though naturally available biopolymers are the most abundant, synthetic biodegradable polymers have been designed and synthesized to get the required physical and chemical properties finding varied applications in cosmetics, coatings, wound dressings, enzyme immobilization, gene delivery, and tissue engineering scaffolds. Further, structurally-modified biopolymers are more prescribed and suggested for their usage in nanotechnology, augmentation, cryopreservation, prosthetics, medical implants, devices, and sanitation products including surgical sutures. It is important to understand the structural properties of various biodegradable polymers for their applications. In this regard, the present review article describes the classification of bio-degradable polymers as natural (polysaccharides and proteins) and synthetic (esters, amides, ethers, urethanes, or hybrid systems) biodegradable polymers by describing their structural properties suitable to the biomedical applications. The development of new hybrid biodegradable polymers with flexibility and long-term biocompatibility leads to the flow of numerous non-toxic biomaterials appropriate to biomedical fields like orthopedics, long-term implants, regenerative medicine, and controlled drug delivery experiencing vital contributions. The enormous efforts and investigations made by these biodegradable polymers facilitate the significant expansion of the development of new viable products for clinical trials.

Abbreviations: PLA, Poly(lactic acid); PBS, Poly(butylene succinate); PCL, Poly(caprolactone); MCF-7, Michigan cancer foundation-7; PBM, Photo-biomodulation; PLGA, Poly(lactic-co-glycolic acid); POF, Polymer optical fiber; ECDs, Endocrine-disrupting chemicals; PEG, Polyethylene glycol; CH, Chitosan; CNF, Cellulose nanofiber; ROP, Ring-opening polymerization; PGA, Poly(glycolic acid); HSAC, α-Hydroxysulfonic acid cellulose; PHAs, Poly(hydroxy alkanoates); CPR, Composite polymer resin; PVA, Polyvinyl alcohol; SEM, Scanning electron microscope; SD, Standard deviation; CV, Coefficient of variation; GNPs, Graphene nano-platelets; PBAT, Poly(butylene adipate-co-terephthalate); PRP, Polymer-reinforced paper; PLLA, Poly(L-lactic acid); PHBV, Poly(3-hydroxybutyrate-co-3-hydroxy valerate); BP-DES, Biodegradable-polymer drug-eluting stents; DP-DES, Durable-polymer drug-eluting stents; BMS, Bare metal Stents; VLST, Very late stent thrombosis; UTS, Ultimate tensile strength; BDPF, Biodegradable polymeric fiber; PET, Poly(ethylene terephthalate); PVC, Poly(vinyl chloride); PS, Poly(styrene); BTX, Benzene-toluene-xylene; MDI, Methylene diphenyl diisocyanate; BDO, 1,4- Butane diol; PTMO, Poly(tetramethylene oxide); IPDI, Isophorone diisocyanate; DMPA, Dimethylol propionic acid; PU, Poly(urethane); TEA, Triethylamine; TDI, Toluene diisocyanate; HBCS-C, Catechol hydroxybutyl chitosanl; DOPA, 3,4-Dihydroxy phenyl-L-alanine; BC, Bacterial Cellulose; SM, Streptomycin; GEN, Gentamicin; BCP, Bacterial cytological profiling; BHK, Baby hamster kidney fibroblast cell; MRI, Magnetic resonance imaging; SMP, Shape memory polymer; SMM, Shape memory material; BSMP, Biodegradable shape memory polymer; CMGG-g-PEI, Carboxy methylated guar gum grafted poly(ethyleneimine); LMW-b-PEI, Low molecular weight branched poly(ethylenimines); BTCA, 1,2,3,4-butanetetracarboxylicdianhydride; GG, Guar gum; LPEI, Low molecular polyethyleneimine; FDCA, 2,5-furan dicarboxylic acid; CAN, Covalent adaptive network; HSAC, Hydroxy sulfonic acid cellulose; M_t, Amount of water absorbed; D_{eff}, Diffusion coefficient; TMDI, 2,4,4 trimethyl hexamethylene diisocyanate; HDI, Hexamethylene diisocyanate; PHDI, 1,10-phenanthroline-5,6-dione; BD, 1,4-Butanediol; BIN, N,N-Bis(2-hydroxylethyl)isonicotinamine; HPED, N,N,N',N'-Tetrakis(2-hydoxypropyl)ethylenediamine; PETA, Pentaerythritol tetraacrylate; AA, Acrylic acid; AN, Acrylonitrile; AT, Aniline trimer; BA, *n*-Butyl acrylate; BAC, N,N′-bis(acryloyl)cystamine; PEG, Polyethylene Glycol; PDMS, Polydimethylsiloxane; PLCL, Polycaprolactone; PSA, Pressure-sensitive Adhesive; PDLLA, Poly(DL-lactide); POPD, Poly(*o*-phenylenediamine); ODX, Oligo(p-dioxanone)diol; PPC, Polypropylene Carbonate; TMC, Trimethylene carbonate; GA, Gum Arabic; CL, Cross linked; PGS, Poly(glycerol sebacate); PGD, Poly(glycerol-dodecanoate).

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https://doi.org/10.1016/j.eurpolymj.2023.112068
Received 15 January 2023; Received in revised form 5 April 2023; Accepted 7 April 2023
Available online 10 April 2023
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1. Introduction

Because of the production of huge quantities and the excessive usage of synthetic plastics for our comfort and convenience, there has been a serious concern in the recent past about the high toxicity and non-biodegradability. This has led to the inevitable ban on plastics meant for single use in 2021 [1–4]. Now, the importance of 3D printing is gaining a lot of attention and the consequences of this new technology and its products can only be seen shortly [3,4]. The current pollution problem is being caused by: (i) the inappropriate disposal of plastics by consumers/industries due to either the lack of awareness about the consequences or inefficient waste collection infrastructure and (ii) the high chemical stability of the plastics [5].

Polymers or plastic objects have become a part of the present society because they possess several properties [6]: lightweight, tunable strength and elasticity, easy shaping and coloring, tunable transparency, thermal/electric insulation and many more. These properties are rewardingly exploited for a variety of applications. Some include rigid containers and castings, flexible films for packaging, rigid sheets for finished two-dimensional articles, fibers for textiles and ropes, foams for cushions and mattresses or all kinds of surface coverings, shock-absorbing joints, or assembling agents such as coatings, adhesives, sealants, and elastomers. These various applications require different combinations of properties. Particularly demanding applications can further be reached by reinforcing polymers with strong materials such as glass or carbon fibers, thereby forming composites.

In general, polymers can be broadly classified into natural (polysaccharides and proteins), synthetic (esters, amides, ethers, urethanes) biodegradable polymers, and synthetic-biopolymers (or hybrid systems) (Fig. 1).

The decomposition of organic materials by microorganisms like fungi and bacteria is known as biodegradation and this is different from composting in which the ideal conditions are created for biodegradation to take place as shown in Fig. 2 [7]. Though many materials are biodegradable, plastics and glass might take thousands of years to break down [8]. In general, biodegradation occurs through biodeterioration as a first step in which the disintegration of physical, chemical, and mechanical properties of a polymer happens with abiotic components in the open environment. Secondly, the fragmentation process proceeds through the bond-breaking of polymeric species into monomers and oligomers [9]. Aerobes are aerobic bacteria that efficiently break down a variety of organic contaminants in an oxygen-rich (aerobic) environment [10,11]. On the other hand, anaerobic digestion occurs more slowly and over longer periods in the absence of oxygen [12–15]. Furthermore, the byproducts of fragmentation are subsequently incorporated through the membrane carriers into microbial cells during the next step *i.e.,* assimilation [9]. As soon as the products are within the cell, they enter catabolic pathways, which either result in the generation of adenosine triphosphate (ATP) or structural components for the cell [9].

With the worst consequences of highly stable plastic objects to the world, biodegradable polymers have been proposed for the replacement of many plastics which are currently being used extensively. But this is not possible straightaway because biodegradation depends on many factors such as density, surface area, pH, air, light, temperature, and the nature of the microorganisms are always considered a complex process [16]. At present, biodegradation is being utilized in various processes *i.e.,* from industrial composting to home composting, but sea water and landfill are the least favorable ones. These biodegradable polymers break down into smaller chemical fragments under oxidative degradable conditions, unlike stubborn synthetic polymers or plastics. Although the biodegradation of polymers has been explored extensively, little research has been carried out on how well such polymers degrade in seawater. But this is a serious issue as the plastic residue settled under the sea water can kill multiple important species. Nakayama et al. examined the biodegradability of polyester through the BOD (biological oxygen demand) method in seawater. As per the report, the biodegradation of polyester was not influenced by tides, but was affected by the temperature of seawater and microorganisms present in seawater [16].

Notably, there are reports available on the mass loss of a class of biodegradable plastics, (polyhydroxyalkanoates, PHAs) in the natural marine environment. The mean rate of biodegradation was found to be 0.04–0.09 mg day^{-1} cm^{-2} ($p = 0.05$), *i.e.,* a PHA water bottle may take 1.5 to 3.5 years for complete degradation. But the research on other marine biodegradable plastics concerning the time and rate of biodegradation is very limited [17].

The chemical composition of these polymers has a significant impact on the options that are available to responsibly manage plastic waste. For instance, most vinyl polymers are persistent pollutants when released into the environment. When collected, however, they are generally good candidates for chemical recycling to hydrocarbon feedstock, eventually via cracking or other stringent chemical modifications. Condensation polymers are more readily susceptible to chemical

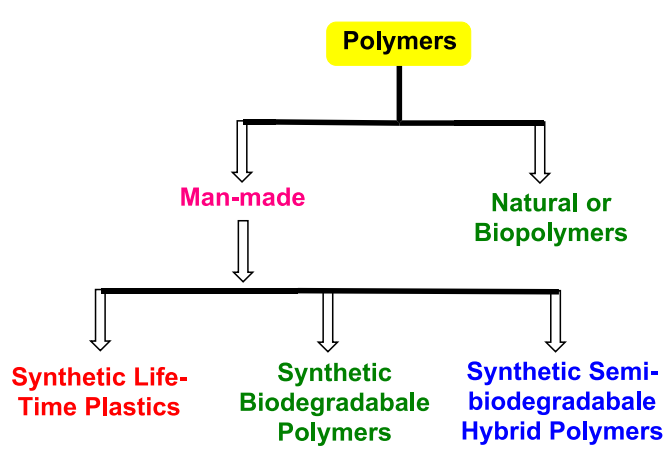


Fig. 1. General classification of polymers.

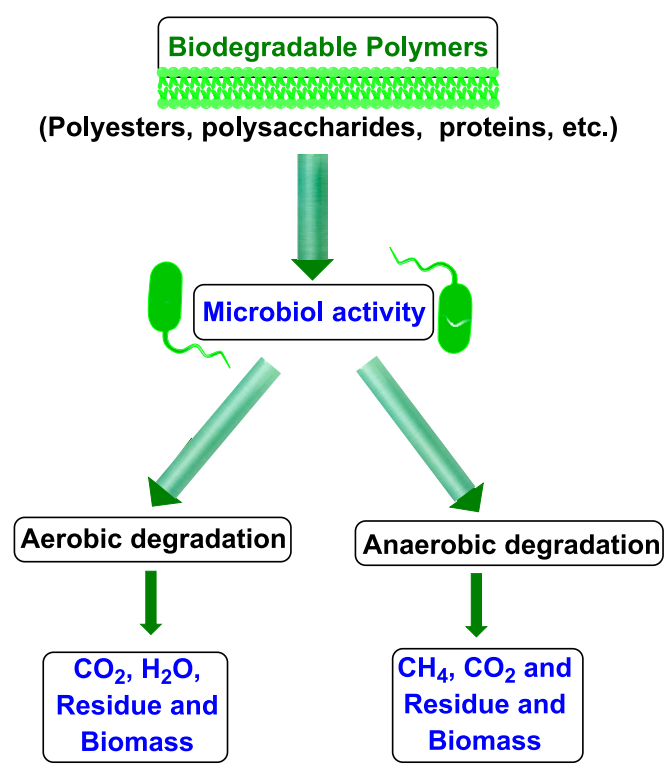


Fig. 2. General mechanism involved in the biodegradation of polymers.

depolymerization: their constituting monomers can directly be recuperated after solvolysis with for instance alcohols or even water. Moreover, these polymers are often also more prone to biodegradation, depending on the type of heteroatom linkages and the concerned monomers. For instance, PET with aromatic flanked ester bonds degrades several orders of magnitude slower in natural conditions (and could be called persistent) than linear aliphatic polyesters such as poly (butylene succinate) (PBS). Condensation polymers are essential in all biodegradable plastic developments.

The biodegradation proceeds in three major steps as described by Rao and co-workers [6,18]: (i) initially different types of wild microorganisms form a biofilm on the plastic surfaces, (ii) subsequent release of a variety of enzymes leads to the de-polymerization of plastics to low-molecular-weight species, and (iii) these components (as a feed) are absorbed by the microorganisms. Due to their high hydrophobicity and the presence of strong C–C bonds, most of the polymers containing aromatic and polyolefin moieties cannot interact with microorganisms to form biofilms and therefore they become resistant to the first two steps of biodegradation. In contrast, hydrophilicity and easily cleavable C-O and C-N bonds make the condensation polymers (*e.g.,* polyamides, polyesters, and polysaccharides) undergo facile biodegradation through the enzymatic hydrolysis by the formation of biofilm. The enzymatic hydrolysis towards the degradation of plastics can be strongly influenced by several other parameters like crystallinity, pH, temperature and the presence of hemiketal bonds in saccharides, aliphatic and aromatic esters, etc. Also, suitable oxidants or photocatalysts aid to reduce the intrinsic resistance of polyolefins. The development of such appropriate tunable oxidants/ catalysts is highly required to enhance the rate of degradation of plastics. Remarkably, the degradation mechanisms and suitable microorganisms are not reported yet for some biopolymers like natural polyamide silk, which is still a challenge similar to stubborn plastics. To reduce the long life of plastics including fossil-based polymers [6,18] and their consequences, thorough research on the usage of the precursors, synthesis, structural aspects, their chemistry, and the plastics should be designed in such a way that they can be degraded easily under selective conditions. This minimizes significantly the accumulation of unwanted plastic waste.

Biopolymers can be used in medical fields like surgical sutures, medical implants, cryopreservation, wound dressings, enzyme immobilization, tissue regeneration, gene delivery, controlled drug delivery, and tissue engineering scaffold to many nanotechnologies with many tunable properties and controllable activities. Biodegradable polymers having conductive properties can be used to directionally control axonal growth and Schwann cell migration by using their microstructures. Using the electronic stimulation of the conducting biodegradable fiber, it is possible to increase nerve growth and Schwann cell migration [19]. For getting long-lasting plastic that will not be harmful to the environment, attention has been paid to poly(lactic acid) (PLA) and poly (butylene succinate) (PBS) (Fig. 3). All these materials can also be used as potential reinforcement materials [20].

Because of the rapidly increasing population in today's world, the demand for food, energy, and resources also increases. Due to the increased population environmental pollution also increases significantly. Now as a possible solution to the water-disposal problem, biodegradable fibers can be produced with limited resources [21].

The biodegradable polymers also have a wonderful property by which they can be biodegradable under seawater also. By doing the alkali treatment of poly(butylene succinate) (PBS) the study of biodegradability was done under the seawater. Because of the alkali treatment, the initial strength decreases first, then after adding starch the biodegradability increases. Here the starch is acting as a decomposition-accelerating substance which makes the spinning of the fiber difficult when added [22]. Not only help the environment by being a solution for pollution but also biodegradable polymers can give an important contribution to the aerospace, automobile, and marine field. This is due to their properties of having high specific strength, and better corrosion and fatigue properties than other materials [23].

The recycling methods of PLA wastes are really difficult and expensive. By drying properly of the recycled PLA flakes and by doing the melt operation the structural properties of the PLA fibers can be controlled [24]. In therapeutic devices also, biodegradable polymers are used as thin and elongated fibers. It is the main component of therapeutic devices. To dress a fresh wound or for surgical or imaging probes, these polymers can also be used. The micro-structured biodegradable fibers can also be used in drug-controlled delivery as a thin film [25]. Poly(lactide) (PLA) (Fig. 4) was found to degrade within a few years under ambient conditions, while PET requires more than 100 years.

Biodegradable polymers are in high demand in tissue engineering including optical-neural interfacing. Tissue engineering is a technique where tissues evolve rapidly to repair, replace or regenerate tissues or organs. The poly(caprolactone) (PCL) (Fig. 5) is taken into different geometries and cross sections like solid-core, hollow-core and grooved fibers for tissue engineering. The cell growth property is shown by the solid-core fibers of different geometries. This cell growth is done by MCF-7 and also the techniques are completely non-toxic [26].

Besides the tissue engineering property, these polymers have bone-regeneration property also. An implantable PBM (photo-biomodulation) technique based on a green light and PLA and glycolic acid to form poly(lactic-co-glycolic acid) (Scheme 1) optical fibers are used in this regeneration process. In a defective bone when a green light is introduced by using fibers, the regeneration and repairing process is accelerated. Biodegradable polymers have been employed to fabricate 3D tissue engineering scaffolds as these materials provide signals to the cells which are related physically and biochemically [27]. Attempts have been made to control the degradation of PLA by the copolymerization of lactide (LA) with other monomers susceptible to hydrolysis. Also, the hydrolytic degradation of co-polymerized PLA with salicylate and cyclic phosphoester enhances significantly [28].

Apart from the tissue engineering property of biodegradable fibers, these polymers can also be used as flame retardants. The flame retardant helps to reduce the intensity of the fire. Alginate salt fibers can also be prepared, which are also biodegradable polymers, by using the wet spinning technique which can be used for cloth materials and flame retardants [29]. The cold drawing affects the various mechanical properties of biodegradable polymers. It improves the tensile strength of the fibers compared with the hot-spun fibers. By increasing the draw ratio an increment in the elastic modulus can be seen [30].

The combed ring-spun yarns can be created from biodegradable fibers. The fibers Modal and Tencel have been derived from the combination of synthetic and blends (50–50 %) of cotton and polyesters for the

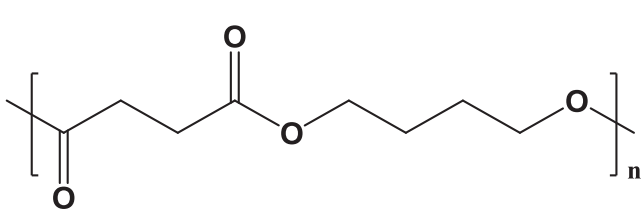


Fig. 3. Structure of Poly(butylene succinate).

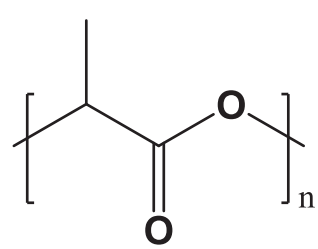


Fig. 4. Structure of Poly(lactic acid).

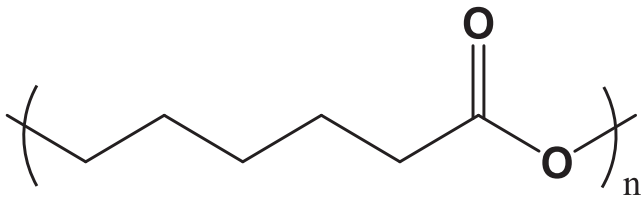
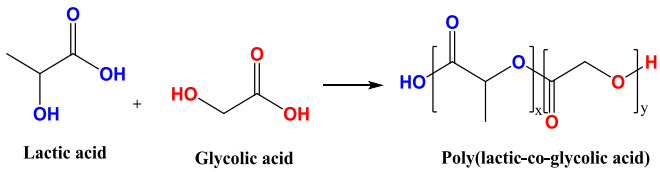


Fig. 5. Structure of Poly(caprolactone).



Scheme 1. Preparation of poly(lactic-co-glycolic acid).

yarn applications [31]. Using biodegradable fiber, a rechargeable fiber battery has been proposed based on the conducting behavior of the fibers using polypyrrole or polydopamine material as a working anode, and MnO_2 acts as a cathode in the presence of biodegradable chitosan and the body fluid as the electrolytes. This gives a capacity of 25.6 mAh g^{-1} with a retention of 69.1 % [32].

Biodegradable fibers can be a potential substitute for petroleum-based plastics, which already have some disadvantages. The main disadvantage is that the sources are limited, also the decomposition rate is very slow. There are few plants in Malaysia and Asia, by which biodegradable fibers can be harvested [33]. This type of fibrous material can be prepared by bamboo-derived nano and microfiber [34]. To reduce the amount of thickening agent used to ensure the proppant carrying capability, biodegradable fiber to the fracturing fluid can be added [35]. As a biodegradable anti-bacterial wound-dressing material, gelatin fibers loaded with ZnO particles can be employed through an electrospinning mechanism. The ZnO particles are uniformly arranged on the gelatin fibers' surface and they don't exhibit any kind of cytotoxicity. This combination is showing a promising effect on the bacterial surface and it is reducing their activity to more than 90 %. Against Escherichia coli (*E. coli*) and Staphylococcus aureus (*S. aureus*), the properties were studied [36].

Further, the incorporation of water-soluble drugs and proteins loaded on microparticles of porous calcium carbonate ($CaCO_3$) into PLGA (poly(lactic acid-co-glycolic acid)) fiber-based scaffolds using co-precipitation and electrospinning methods was described. The presence of $CaCO_3$ microparticles facilitates the second barrier to prevent the burst release of the drug. It was found that the drug release profile was seriously influenced by the pH of the medium employed, drug loading, and drug properties. The drug-releasing rate was observed in the range of 19–30 % in 40 days [37]. By using biodegradable fiber like PLA, many environmental concerns can be minimized and it can also be used as a potential petrochemical substitute or it can be used in numerous applications due to its recyclable property, low toxicity, high mechanical strength and friendly processing properties [38].

The naturally available and one of the most abundant polymers is cellulose which can also be a substitute for petroleum-based plastics because of its convertible power to a biodegradable and non-toxic substance. After converting, it can be used as coating materials, paper boards for food packaging, etc. [39] One of the most important contributions of biodegradable fibers in human life is unconventional energy. There are several sources of unconventional energy. They are wind, tidal, solar, nuclear, etc. They are renewable sources. To get electrical power from kinetic energy, a wind turbine can be used. The blades, gearbox, and even the wind turbine blades are made up of biodegradable fibers as they have high mechanical strength, low weight and better corrosion resistance. [40].

Biodegradable polymer optical fiber (POF) is also a biodegradable fiber that can be used in many medical applications. In drug delivery, neutral recording, biosensing and optogenetics POF are used [41]. Biodegradable fibers can also be used to remove endocrine-disrupting chemicals (EDCs) from wastewater. These EDCs like industrial compounds, dyes, pharmaceutical wastes, etc affect the natural hormones in human and animal bodies. These chemicals may lead to cancer, obesity, autism, thyroid problems, etc. So, for the removal of EDCs from the wastewater, Biodegradable fibers work as an adsorbent [42]. The usage of diester monomers (glycolide and lactide) produces biodegradable polymers having high molecular weight. Fibers of these monomers are mostly used in pharmaceutical industries or food industries or additive technologies [43]. By using the click chemistry method, biodegradable PEG polyester copolymers can be produced. At first, *m*PEG was employed as the initiator and then a ring-opening polymerization was done by using Sn $(Oct)_2$ as a catalyst to the α-alkynyl functionalized lactone portion with the D, L-lactic acid. Further, these PEG poly(esters) (Fig. 6) can be used in drug delivery industries [44].

The chitosan polymer (CH) is known for its excellent physical and structural properties with its activity against bacteria. Though this polymer has some drawbacks that can be modified by some chemical modifications. By doing the graft or cross-linking modification, the properties like solubility, biological activity, mechanical properties, etc. can be improved [45]. Cellulose nanofiber (CNF) is also a famous biodegradable fiber that can also be used as a reinforcement material because of its excellent mechanical properties, low density, promising bio-compatible properties, and tensile strength. This can be prepared from silkworm silk fibers by including the CNF in the diet of a silkworm [46].

Biodegradable polymers require an appropriate strength, similar to plastics for applications like building materials, hygiene products, and packaging and commodities to reduce waste-disposal problems like reduced resistance to microbial degradation when they are in an ambient environment. Biodegradable plastics have been obtained from both synthetic (*i.e.*, non-renewable petroleum resources) and natural polymers (*i.e.,* renewable sources) [47–49]. Biodegradability strongly depends on the source of the polymers, their chemical structures, and the environmental conditions including the mechanistic pathways [50]. The mechanical applications of biodegradable polymers are generally influenced by their composition [51,52], processing characteristics [53,54] and aging conditions. [55] Biodegradable materials can be classified into three groups: (i) the blends of manufactured polymers that are edible by microorganisms (starch and cellulose), (ii) synthetic materials possessing a group undergoing hydrolytic microbial attack (poly(caprolactone), and (iii) the bio-polyesters derived from bacterial sources. Notably, the aliphatic polyesters like poly(ε-caprolactone), polylactide, poly(3- hydroxybutyrate), poly(ethylene oxide), poly(glycolic acid), and thermoplastic protein were found as predominant and vital biodegradable polymers [56]. The biodegradable plastics can be composted with organic wastes to improve soil quality and other related applications [57].

The selected salient features of biodegradable biomaterials are: (a) they possess a time of degradation according to their functionality; (b) they don't contain a sustained inflammatory response; (c) they do contain suitable permeability for designed applications; (d) they do contain proper application-based mechanical properties, and finally, (e) nontoxic degradation products produce during the biodegradation process. [58].

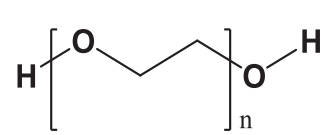


Fig. 6. Structure of Poly(ethylene glycol).

2. Classification of biodegradable polymers:

- Natural biodegradable polymers – *e.g.,* poly(saccharides), lignin, chitosan, starch, cellulose, guar gum, collagen, albumin, etc [59,60].
- Synthetic biodegradable polymers - *e.g.,* aliphatic polyesters, poly (orthoesters), poly(anhydrides), poly(phosphazenes), poly(amino acids), vitrimers, BDMPs, etc [61].

2.1. Natural biodegradable polymers:

Natural biodegradable polymers are an important class of biodegradable polymers. The basic features of natural biodegradable polymers are that they are easily available and derived from natural sources and relatively inexpensive *e.g.,* polysaccharides, Lignin, chitosan, starch, cellulose, guar gum, collagen, albumin, etc. [62] as shown in Fig. 7.

2.1.1. Polysaccharides

Apart from polyesters, the most traditional and abundant biodegradable polymers are polysaccharides (Fig. 8) [63]. The regenerated cellulose materials and starches have been widely employed as one of the major components in packaging materials. The presence of free hydroxyl groups in polysaccharide materials can be easily softened for the applications *e.g.*, xanthate groups (–O–(S)C–S–)/ ester (–O–(O)C–) in the presence of glycerol [64,65]. In the recent past, a variety of hybrid-polysaccharide materials have been prepared based on microbial sources. In addition, Molenveld and Slaghek reported a class of cellulose-based thermoplastics derived from the esterification of three available –OH groups (per mole of glucose entity) possessing excellent optical and mechanical properties [66]. Furthermore, the development of starch alternative materials produced naturally by microorganisms instead of plants was very well-reviewed by Iwata and coworkers [67]. These interesting microbial polysaccharides can be connected with glucose through 1,3-, 1,4- or 1,6-glucosidic bonds in equatorial (α)/ axial (β) configuration than the cellulose and starch produced by the plants [66,67], In addition to the free hydroxyls of these microbial polysaccharides, other functional groups allow the production of plastics similar to polyolefin, adhesives, and high-performance-based engineering plastics with selective optical transparency. These modifications

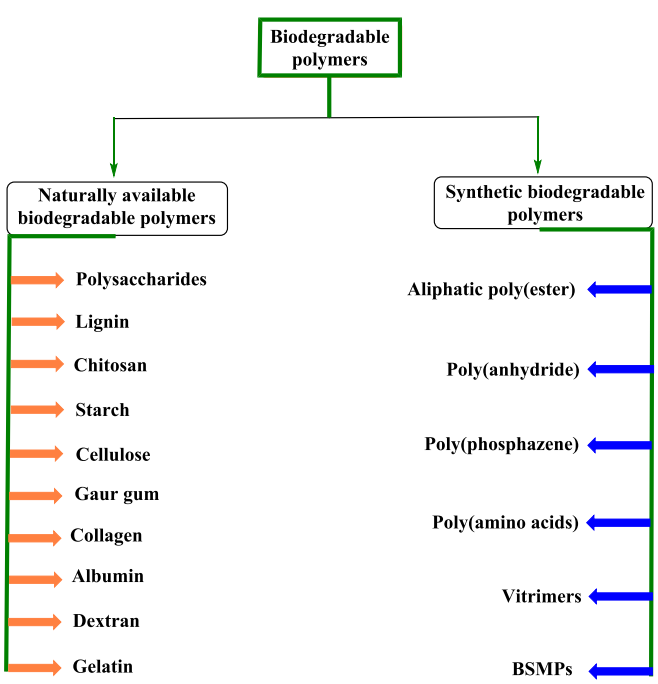


Fig. 7. Classification of biodegradable polymers based on their availability.

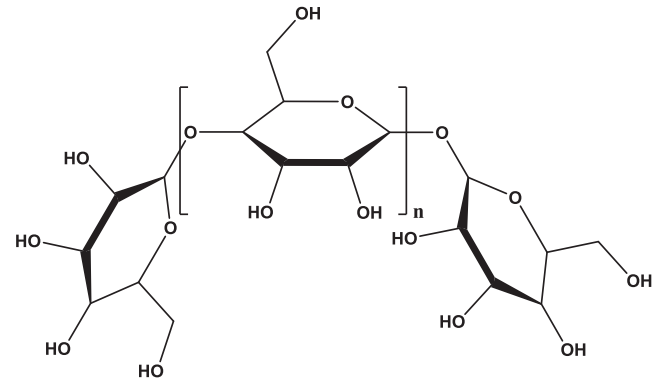


Fig. 8. Structure of Polysaccharides.

influence the biodegradability of the poly(saccharides) in terms of their affinity towards the formation of biofilms and further accessibility of the hydrolyzable hemiketal/glycosidic bonds.

2.1.2. Lignin

After polysaccharides, lignin is the second most abundant biopolymer on earth. This always encourages scientists to focus research on the development of upgraded materials from lignin (*e.g.,* biodegradable thermoplastics) [68–70]. This is done primarily by fixing the functional groups to the lignin structure facilitating cross-linking, or employing lignin instead of phenol in phenol–formaldehyde wood glue [71]. According to Sarkanen and Chen [71], lignin can also be a linear phenolic polymer of ~2 kDa forming complexes of ~20 kDa in alkaline water and 2000 kDa in polar organic solvents reversibly after derivatization apart from the formation of highly cross-linked materials. This leads to the production of materials by derivatizing lignins in the presence of minor amounts of monomeric or polymeric components with an enhanced tensile strength equivalent to poly(styrene) (PS). As lignin (Fig. 9) is one of the most biodegradable materials with microorganisms through strong oxidation and radical-forming enzymes (*e.g.,* laccases), the development of fully biodegradable lignin materials opens new platforms for promising applications. In addition to this, different types of substituted aromatic moieties in lignin can be linked to the important precursors for many high-performance materials like polycarbonates, polyesters, polyamides, epoxy-resins, renewable bisphenol alternatives, BTX (benzene–toluene–xylene), etc [72].

Lignin can be used to create biofuels like bioethanol, biogas, and bio-oil. The removal of lignin from the plant material and subsequent conversion of the remaining hemicellulose and cellulose into biofuels is referred to as lignocellulosic biofuel production. [73] Lignin is a raw material that can be used to create a variety of compounds, including

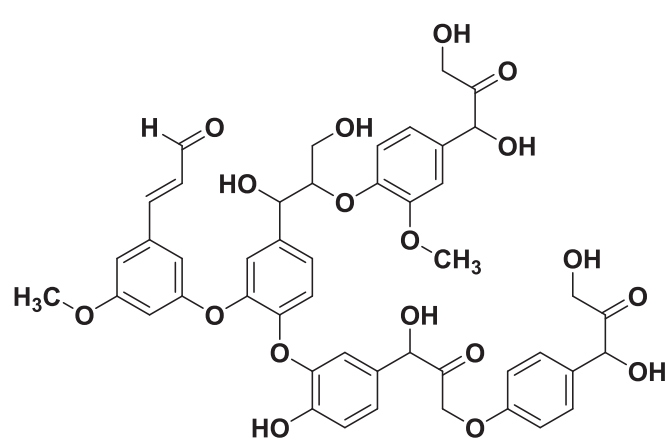


Fig. 9. Structure of Lignin.

phenol, which is used to make polymers, resins, and adhesives, and vanillin, a flavoring ingredient. Carbon fibers, which are utilized in the aerospace sector, can also be made from lignin [74]. It can be utilized as a natural insecticide and soil conditioner. It enhances soil structure, holds onto moisture, and gives plants nutrition when put into the soil. As a natural insecticide, lignin can be used to manage pests and illnesses that are spread through soil [75]. Lignin can be utilized to create sustainable and biodegradable polymers. It can replace petroleum-based plastics by acting as a filler or reinforcement material in the manufacturing of bioplastics [76]. It can be used to make fibers for the industry of textiles. Strong, long-lasting, and moisture-wicking, lignin-based fibers are ideal for use in home textiles, outdoor gear, and athletic equipment [77].

2.1.3. Chitosan

Functionalizing synthetic polymers with natural ones, especially polysaccharides like chitin and chitosan, can speed up the development of biomaterials. Chitin and chitosan-based polyurethanes, hydrogel nanoparticles, bio-adhesives and biomaterials offer tremendous biological properties across a wide range of applications, making them a potentially more interesting option (Fig. 10).

The reaction of 4,4-diphenylmethane diisocyanate (MDI) and poly (ε-caprolactone) (PCL) produce different derivatives of chitin-based poly (urethane) elastomers (Fig. 11) at varied mass ratios of 1,4- butanediol (BDO) and chitin towards biomedical applications since they are non-absorbable sutures [78].

Chitin/polyurethane networks and blends were modified by combining them in two different macromolecular configurations (Fig. 12). Initially, the blends are prepared by mixing the polymers, and later, the tridimensional networks are developed for the simulation of living tissue. Additionally, in vitro studies showed that the polyurethane materials do not release any harmful products and possess low Vero cell adhesion [79].

Block polymers of poly(urethane) and chitosan were synthesized and prepared through the poly(urethane) prepolymer derived from PTMO (Mn = 1000), IPDI, and DMPA, and further followed by ionizing PU prepolymer by trimethylamine (TEA). Finally, the polyurethane chain was extended by water-soluble chitosan (Mn = 5000) through an emulsion polymerization technique. These block polymers of polyurethane exhibit good anticoagulant properties but are not good candidates for antibacterial and antifungal properties (Fig. 13) [80].

The chitosan-based poly(urethane) was fabricated by step-growth polymerization using 2,4-toluene diisocyanate (TDI) and poly(ε-caprolactone) (PCL). Then, the polyurethane prepolymer was extended with varied ratios of 1,4-butanediol and (BDO) chitosan and used for molecular engineering (Fig. 14) [81].

The functionalized (HBCS-C) were prepared by grafting hydroxybutyl groups and catechol groups for tissue adhesion and wound hemostasis (Scheme 2) [82].

2.1.3.1. Chitosan/DOPA-Based hydrogel. Chitosan-based nanoparticles have been used in drug and gene delivery, cell imaging, wound healing, antibacterial formulations, restorative dentistry, cancer and hyperthermia therapy and tissue engineering [83]. Chitosan-based bio-adhesives have been explored in wound dressing, sutureless surgery, drug delivery systems, and hemostatic applications [84]. Also, chitosan-based

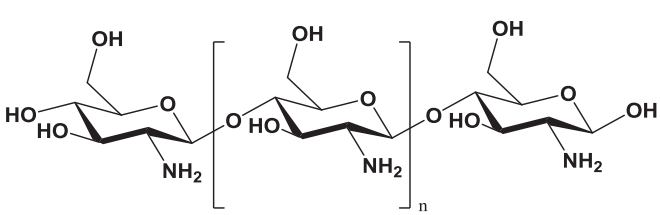


Fig. 10. Structure of Chitosan.

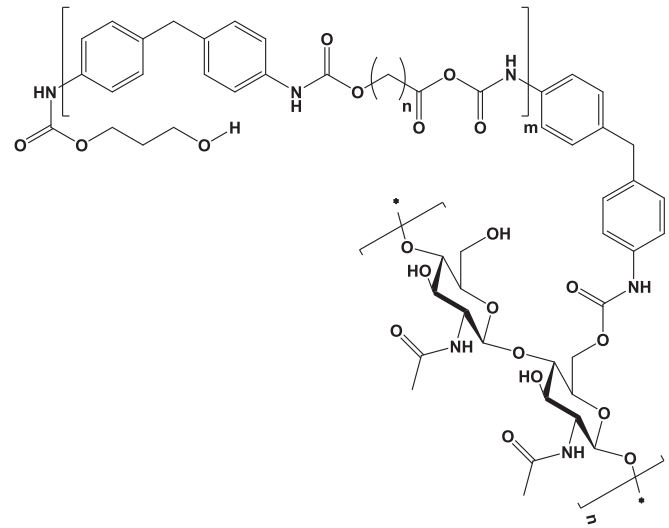


Fig. 11. Structure of chitin/ 1,4-butane diol-based poly(urethane).

biomaterials were used for the treatment of bone disorders [85].

Chitosan has been employed in medical applications such as tissue engineering, drug delivery systems, wound treatment, and medical devices. For these purposes, its biocompatibility and degradability make it the perfect material [86]. In agriculture, chitosan is utilized as a natural insecticide and a stimulant of plant development. It raises crop yields, enhances plant development, and strengthens plant protection against diseases when sprayed on crops [86,87]. Chitosan is utilized as a preservative in the food industry because of its antibacterial qualities. To cut calories, chitosan is sometimes utilized as a fat replacement in food products [88]. Chitosan is employed in the treatment of water to eliminate organic contaminants, heavy metals, and colors from wastewater. It functions as an efficient water treatment agent due to its capacity to create complexes of metal ions and absorb organic molecules [89]. Chitosan is utilized in cosmetics because it can repair skin. To enhance the health of the skin and hair, it is utilized in shampoos, skin creams and other cosmetic items [86,90].

2.1.4. Starch

The carbohydrates reserved in plant tubes and seed endosperm plants are starch, which is found as granules in plastids that are isolated from the cytoplasm. The two main sources of starch are rice and corn. A white, amorphous powder, starch has no sweetness. It is a non-reducing carbohydrate and insoluble in water, alcohol, and ether. Granules of starch can be simple or complicated, and their sizes (2–150 m), size distributions, and shapes can vary. Amylose and amylopectin, two glucans, are combined to form starch. Depending on the source, the majority of starches comprise 80–90% amylopectin and 10–20 % amylose which are water-insoluble. While sugary mutant corn and some legumes have amylose, glutinous or waxy starch from corn and other cereals contain little to no amylase (Fig. 15). [91].

2.1.4.1. Amylose. Amylose is a type of straight-chain polysaccharide in which units of 1–4-D-glucose are linked. Chain lengths range between 250 and 350 glucose units, and the lengthy molecules have the appearance of being coil-like. In water, amylose dissolves, but it also forms hydrated micelles. The lengthy chain is wound into a helical coil in such micelles. The blue color that results from iodine and starch is caused by this structure. The way cooked and cooled starches gel is influenced by the structure of amylase (Fig. 16) [91,92].

2.1.4.2. Amylopectin. Amylopectin also possesses a backbone made up of - (14) connections, but 4–5 percent of the molecule is also branched by - (16) linkages. In amylopectin, the linear unit is 20–25 percent

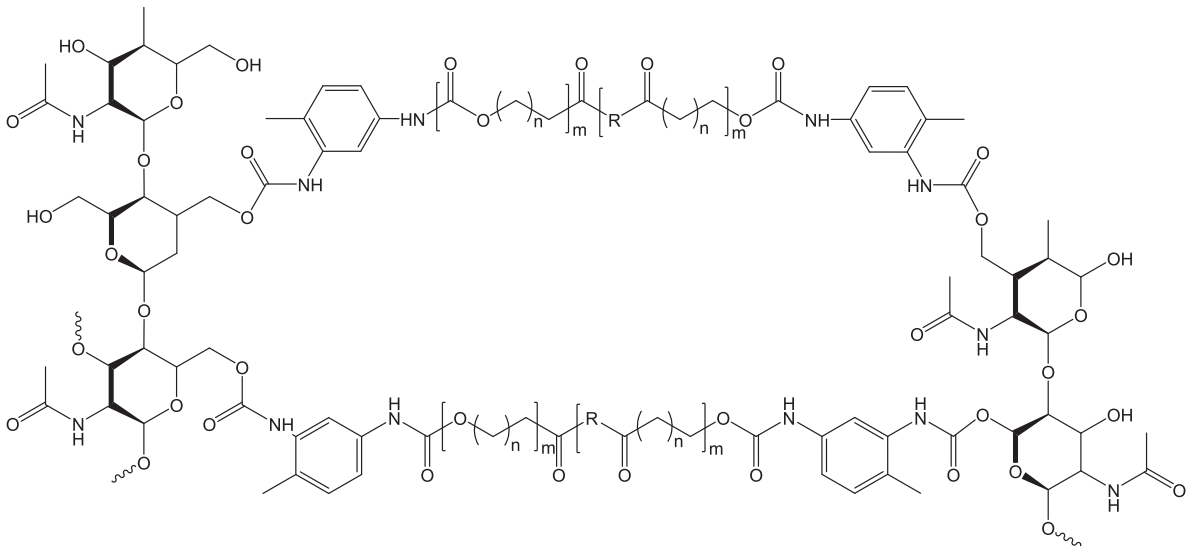


Fig. 12. Structure of chitin/poly(urethane) network and blend.

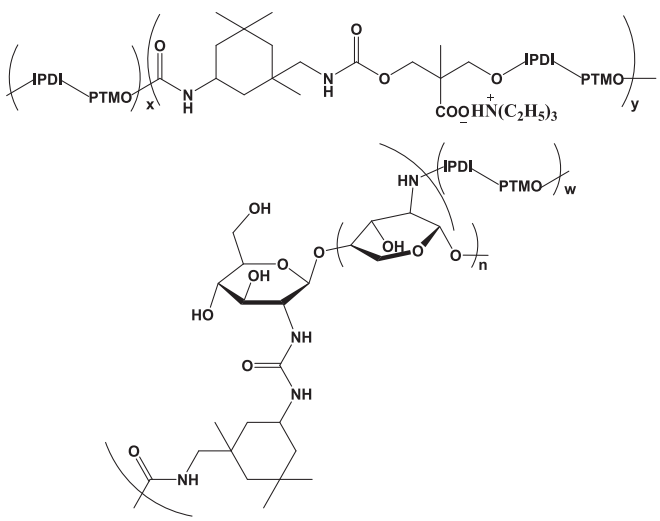


Fig. 13. Synthesis of an emulsion possessing polyurethane urea and chitosan as the chain extender.

glucose units long. The thickening effects of starch preparations are caused by amylopectin, although it does not affect how gels are made. According to X-ray diffraction, starch granules have a semi-crystalline structure, with 30% of them being crystalline and 70% of them being amorphous. The main component of the crystalline regions is amylopectin, while the main component of the amorphous portions is amylase. The starch's moisture content affects how crystalline the substance is. Crystallinity levels are lower in dried starches. Because of the hydrogen bonds created by the –OH group of the granule surface, starch granules are hydrophilic and have strong intermolecular interactions. To alter the starch, chemicals are typically used. However, combining starch with other starches can enhance or change the properties of starch (Fig. 17).

It is crucial to functionalize synthetic polymers with natural ones, especially poly(saccharides) like starch, to improve the development of biomaterials. They offer outstanding biological properties across a variety of applications, making starch-based hydrogel, films, biodegradable materials, copolymers, and electrospinning polymer a possibly more spectacular proposition [91]. Starch-functionalized with thiourea (starch-thiol) was oxidized to sulfonic acid derivative, and then the quaternized starch sulfonic acid derivative was synthesized and copolymerized by redox initiation system with silica nanoparticles at 40 °C to avoid starch gelatinization for potato starch. These starch-thiol derivatives can be employed in chemical flooding technologies for improving oil recovery (Fig. 18) [93]. Starch-based grafted polymers have been employed for tissue engineering, drug delivery, and diagnostic imaging. Various electrospinning fibrous form of starch is used for cytotoxic effect and bone tissue engineering [94].

Biodegradable and green packaging materials like bags, cups, and containers have been made using starch polymers [95]. Starch is employed as the primary raw material in the creation of biodegradable plastics. These plastics are utilized in a variety of products, including consumer items, agricultural films, and food packaging [96]. It is employed as a sizing agent for yarns in the textile industry. They increase the fibers' tensile strength and suppleness and lessen breakage while weaving and spinning [97]. It is employed as coatings and binders in the paper business. They enhance the paper products' printability, strength and surface qualities [98]. Starch polymers are a binding agent used in the creation of adhesives. They have several uses, including labeling, packing, and book binding [99]. Shampoos and conditioners are just a few examples of personal care items that use starch polymers. They serve as stabilizers and thickeners, enhancing the products' viscosity and texture [100].

2.1.5. Cellulose

The most prevalent type of biomass is cellulose globally, which is being used in contemporary industries. The cellulose chains are linked up with sugar and D-glucose. The molecular chain of cellulose extends in a relatively straight line as a result of this arrangement, which places all functional groups at equatorial positions, making it an effective fiber-forming polymer. The hydroxyls on the cellulose chain are positioned equatorially, where they protrude laterally along the stretched molecule and are easily accessible for hydrogen bonding. Numerous elements such as variety, harvest, disintegration (mechanical and steam explosion treatment), climate, retting level, maturity, decortications, textile, fiber modification, and technical processes all have an impact on the qualities of cellulose fibers (spinning and carding) (Fig. 19) [101].

It is possible to hasten the development of biomaterials by functionalizing synthetic polymers with natural ones, particularly polysaccharides like cellulose and its derivatives. Hydrogels based on cellulose, scaffolds, nanoparticles loaded with silver, magnetic nanohybrids, and nanocomposites all have remarkable biological features that make them potentially more appealing opportunities.

Cellulose nanocrystals and cellulose nanofibrils-based hydrogels for

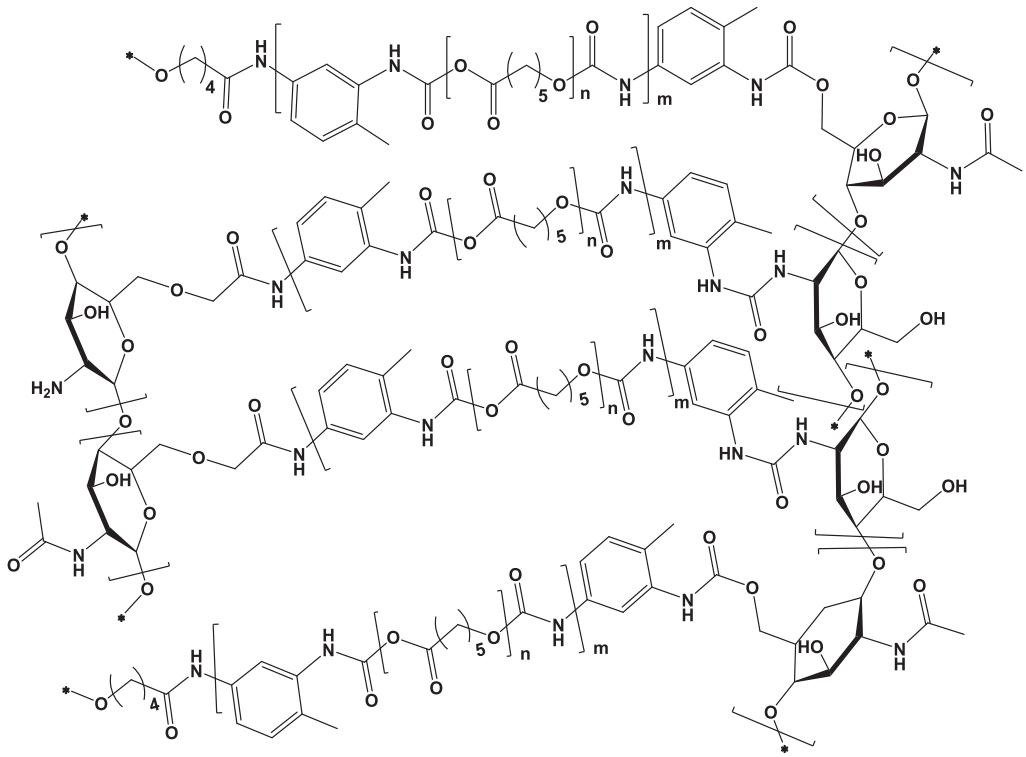
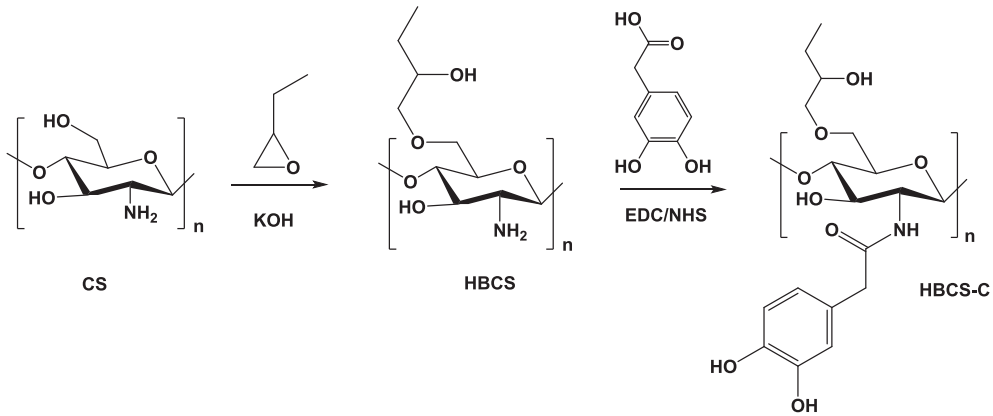


Fig. 14. Chitosan-based polyurethane elastomers using aromatic diisocyanate.



Scheme 2. Preparation of catechol-based hydroxy butyl chitosan hydrogels.

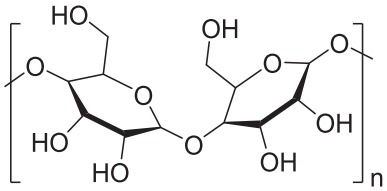


Fig. 15. Structure of Starch.

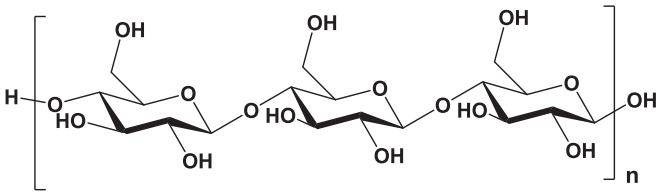


Fig. 16. Structure of Amylose.

wound dressings, drug delivery, and tissue engineering scaffold applications [102]. Composite scaffolds based on bacterial cellulose (BCP-GEN and BCP-SM) were prepared by impregnating polycaprolactone (PCL) into the BC matrix for the fabrication of flexible bacterial cellulose-based PCL membranes (BCP). Then these materials were further functionalized with antibiotics streptomycin (SM) and gentamicin (GEN) separately. The BCP scaffolds exhibited excellent biocompatibility toward baby hamster kidney fibroblast cells (BHK-21) and show good antimicrobial activity as a wound dressing material (Fig. 20) [103].

Green synthesis of bacterial cellulose-based silver-loaded nanoparticles was used for the antimicrobial application [104]. Cellulose-based magnetic nanohybrids were used for MRI, wound healing, stimuli-responsive drug delivery systems, nucleic acid extraction,

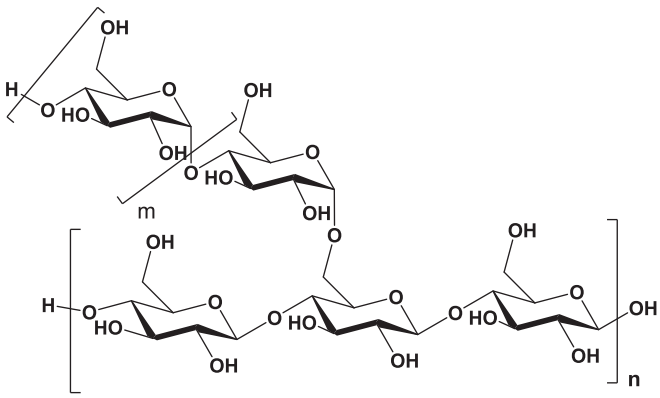


Fig. 17. Structure of Amylopectin.

enzyme encapsulation, and tissue engineering [105]. Similarly, cellulose-based nanocomposites were employed for bone tissue engineering, medical implants, drug carriers, wound healing, and three-dimensional printing [101].

Cellulose is the primary ingredient of paper and paperboard, according to the paper industry. It is utilized as a raw resource to make a variety of paper products, including packaging paper, tissue paper, and newsprint [106]. The production of fibers like cotton and rayon uses cellulose as a basic material. In the making of yarns, it also functions as a sizing agent [107]. It is a bulking agent and food additive used in many different food items. It is added to food products including baked goods and meat products to enhance their texture and boost their fiber content [108]. Cellulose is a binder and disintegrant used in the manufacture of tablets and capsules in the pharmaceutical sector. It enhances the powders' flowability and compressibility, making it simpler to produce tablets and capsules [109]. Items for personal care, including toothpaste, shampoo, and cosmetics, contain cellulose. It thickens, stabilizes, and emulsifies the products while also enhancing their consistency and texture [110]. Cellulose is a raw material used in the manufacture of bioplastics. It is mixed with other natural materials to create environmentally responsible and biodegradable plastics [110].

2.1.6. Gaur gum

Guar gum, commonly known as guaran, is a naturally occurring hydrocolloid that accumulates and is kept in the endosperm of guar plant seeds. It is a macromolecular structure that lies between a linear hydrocolloid (like cellulose) and a supercollider (like amylopectin). The chain of (1 to < 4) glycosidic linked mannose units that make up the carbohydrate's backbone is branched every other unit with a (between 1 and 6) linked galactose. The main chain of guar consists of free hydroxyl groups on the mannose unit or a side chain on the galactose unit. It is an almost odorless white to cream-colored amorphous powder that dissolves in both cold and hot water to produce a translucent, colorless whitish colloidal solution. A variety of foods, cosmetics, and pharmaceuticals have been approved for use with guar gum as an emulsifier, thickener, and stabilizer (Fig. 21) [111].

Furthermore, a carboxy methylated guar gum grafted poly(ethyleneimine) (CMGG-*g*-PEI) copolymer was synthesized by Williamson's ether synthesis and through carbodiimide chemistry. The developed copolymer showed an efficient non-viral vector for gene therapy applications. Although the in vitro study revealed the transfection efficiency of CMGG-*g*-PEI/pDNA complexes was considerably higher than that of LMW b-PEI (800 Da)/pDNA. Additionally, the grafted copolymer displayed much lower cytotoxicity compared to the PEI (Fig. 22) [112].

Guar gum hydrogels were prepared by esterification with 1,2,3,4-butanetetracarboxylic dianhydride (BTCA). The acquired hydrogels are revealed as efficient drug delivery carriers for protein-based drugs (Fig. 23) [113].

Low molecular weight polyethyleneimine conjugated guar gum was prepared by using a coupling agent (*i.e.,* naphthalic anhydride) for an effective receptor-targeted and non-viral gene delivery to triple-negative breast cancer (Fig. 24) [114].

It is frequently employed as a thickening, stabilizing, and emulsifying agent in the food business. To enhance the texture and life span of a wide range of food items like ice cream and sauces, it is utilized [115]. It is employed as an adhesive, disintegrant, and coating agent in the pharmaceutical sector. Due to its capacity to transform into a gel-like material when in water's presence, it is also utilized in the

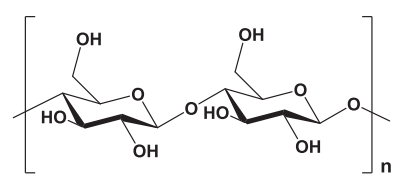


Fig. 19. Structure of Cellulose.

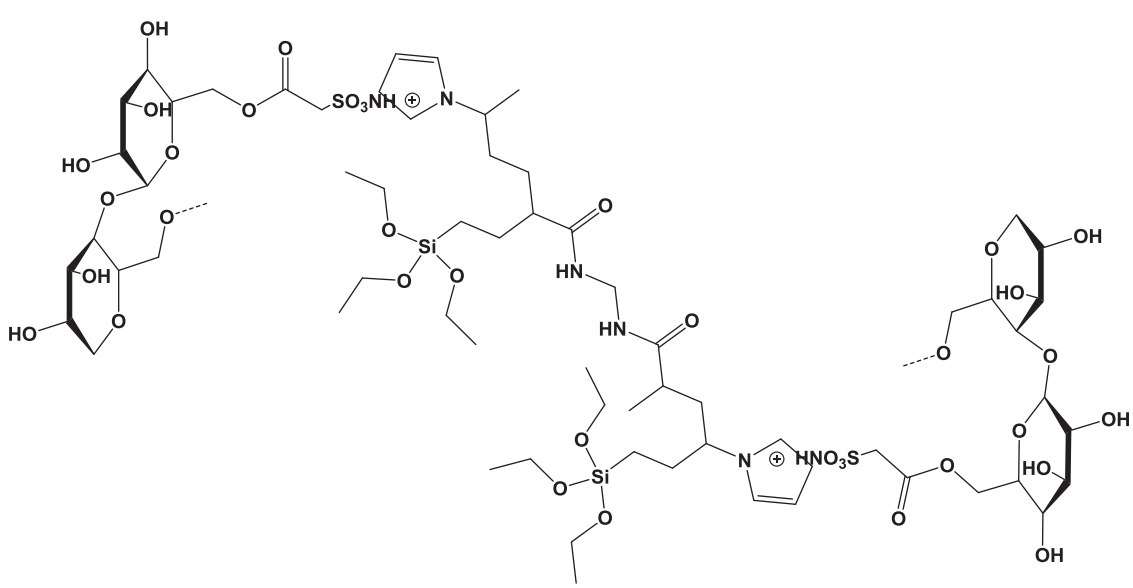


Fig. 18. Starch functionalized sulfonic acid co-imidazolium/silica composite.

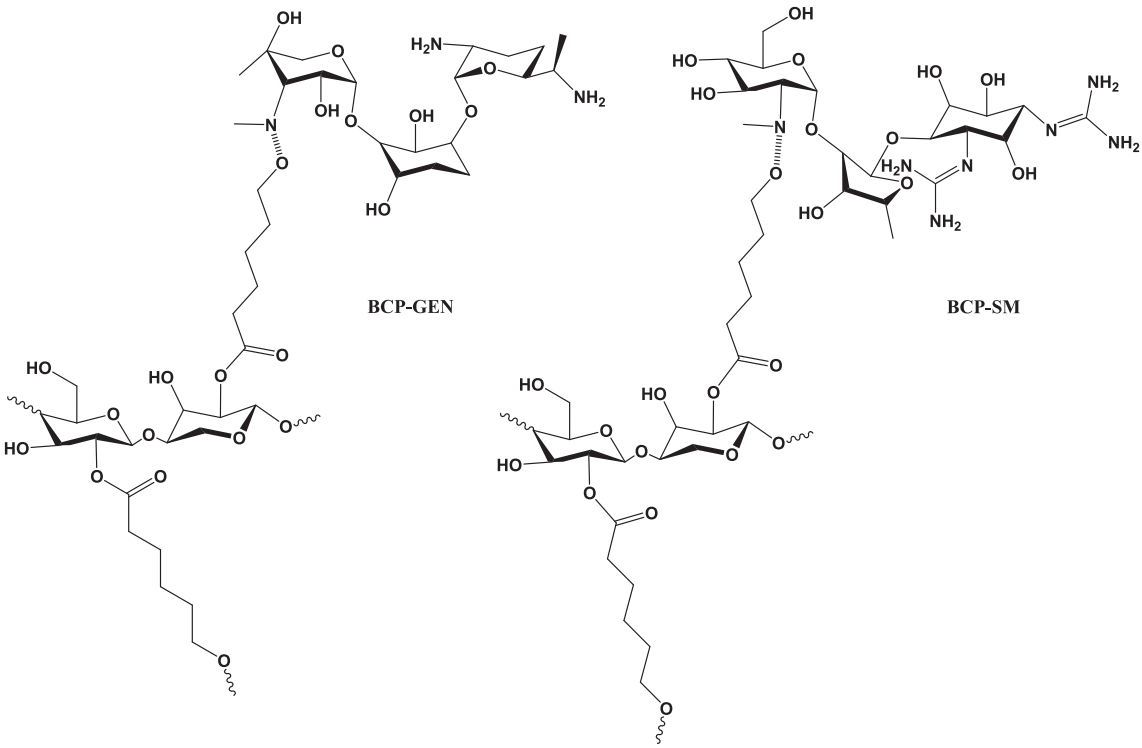


Fig. 20. Impregnation of PCL into the BC membrane and surface functionalization with streptomycin (SM) and gentamicin (GEN).

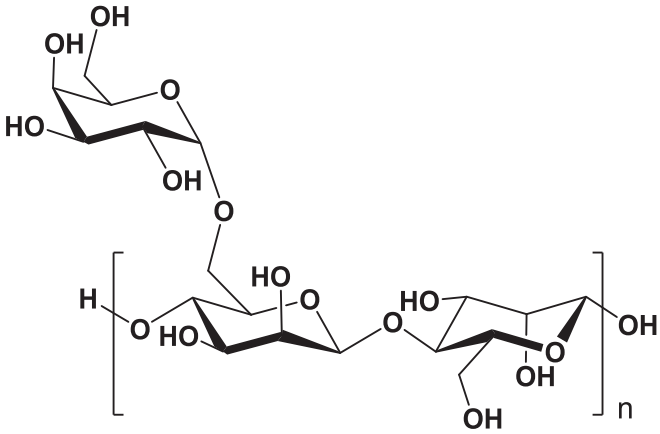


Fig. 21. Structure of Gaur gum.

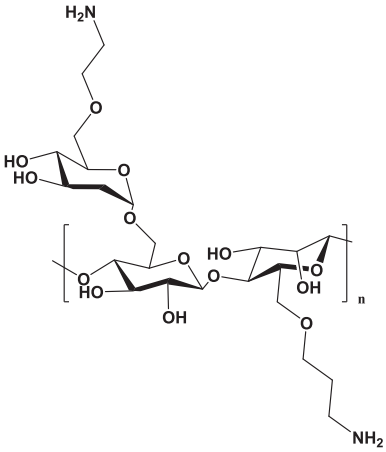


Fig. 22. Carboxy methylated guar gum grafted poly(ethyleneimine) copolymer.

manufacture of drug delivery systems and wound dressings [116]. Guam gum polymer is employed as a sizing agent for fabrics in the textile industry. It gives the materials a rigidity that enhances their general quality [117]. It is utilized as an emulsifier, and stabilizer in the cosmetics sector. To enhance the smoothness and stability of several cosmetic products, including shampoos, lotions, and creams, it is utilized [116]. In the field of biotechnology, It is employed as a substrate for the immobilization of enzymes as well as cells. Moreover, it is employed in the creation of bio-based products like bioplastics. It has been modified and functionalized for use in the delivery of controlled medications [118].

2.1.7. Collagen

Collagen is one of the most widely found proteins in mammals originating in the skin, connective tissue, cartilage, and tendon. It affords structural support to the tissues and this has a very high number of biomedical applications such as tissue engineering of multiple organs, cosmetics, surgery, drug delivery, bioprosthetic implants, sutures, and including dressings (Fig. 25).

Notably, it has reduced dimensional stability with poor mechanical strength and also unpredictable drug release kinetics in drug delivery systems [61,119].

Collagen is employed to enhance the texture and durability of a variety of food products like sausages and other processed meats [120]. In the pharmaceutical sector, collagen is used as a scaffold for tissue engineering, a medication delivery device, and treatment for wounds, burns and skin lesions. Moreover, it serves as a binding agent in the manufacturing of tablets and capsules [121]. Collagen polymer is utilized as an anti-aging component in the cosmetics industry. To increase skin elasticity and lessen the appearance of wrinkles, it is used in many cosmetic items such as creams, lotions, and serums [122]. The collagen polymer is utilized in regenerative medicine and tissue engineering in the creation of scaffolds for the regeneration of bone and cartilage [123].

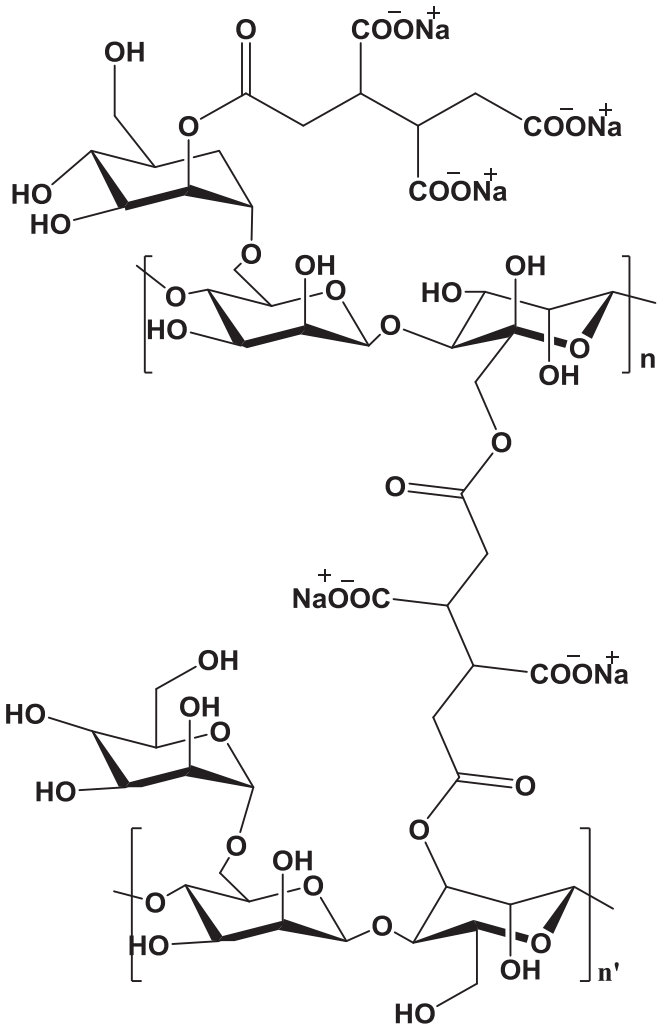


Fig. 23. Synthesis of gaur gum hydrogel *via* crosslinking with BTCA.

Collagen polymer is utilized as a natural substitute for synthetic fibers in the industry of textiles. It is used to make high-performance fabrics for things like athletic gear and clothing for the medical industry [124].

2.1.8. Albumin

Albumin is a family of globular proteins and is available in human plasma as a total protein (>55%). This is mainly used in chemotherapy and designing particulate drug delivery systems for 5-fluorouracil, prednisolone, insulin, sulphadiazine, etc. to achieve high local drug concentration for a long time (Fig. 26) [61,125].

Albumin has been employed as a plasma expander, vaccination adjuvant, and drug delivery device in the pharmaceutical sector. Also, it has been used in the creation of imaging and diagnostic agents [126]. Albumin polymer is used to treat liver illness and as an alternative to blood transfusions [127]. In the food industry, it is used as an emulsifier in many food products including baked foods, whipped toppings, and confectionary items [128]. Various skin and hair care products such as shampoos, conditioners, and face masks made with albumin polymer are utilized in the cosmetic industry [129]. Albumin polymer is a protein stabilizer and additive for culture media in the biotechnology sector. It is employed in the creation of recombinant proteins, vaccines and antibodies [130].

2.1.9. Dextran

Dextran is a complex branched polysaccharide made up of many glucose molecules. Dextran chains are of varying lengths and consist of alpha-D-1, and 6-glucose-linked glucan as side chains linked to the polymer's backbone. The molecular weight of dextran ranges from 1000 to 2,00,000 Daltons. It is used as a drug, especially as a blood plasma volume expander (Fig. 27) [62,131].

Dextran is very useful in the pharmaceutical sector as a vaccine adjuvant, a plasma expander, and a drug delivery method. Moreover, it is employed in the creation of imaging and diagnostic agents [132]. It acts as a thickening and stabilizing ingredient consumed by the food industry. It is included in a wide range of food products, including sauces, dressings, and baked goods [133]. Dextran polymer is extensively applied in the biotechnology sector as a cryo-protectant, protein stabilizer, and additive to culture media and in the creation of recombinant proteins, vaccines, and antibodies [134]. It functions as a scaling agent for cotton in the textile business and provides the materials with the required rigidity that enhances their general quality [135]. In the medical field, it is used to augment the plasma volume in individuals with a low volume of blood. Also, it serves as an alternative to blood transfusions in the treating of thrombosis [136].

2.1.10. Gelatin

Gelatin is a mixture of proteins and peptides obtained by boiling the bones, extraction from the connective tissues, intestines of animals, partial hydrolysis of collagen and organs. It is employed as a coating material and further, the gelatin micro pellets can be used in the oral-controlled delivery of the drugs. The physicochemical properties depend on the source, extraction method and also thermal degradation (Fig. 28) [62,137].

Gelatin is mostly used as a stabilizer, gelling agent, and emulsifier in the food industry to enhance the texture and durability of a variety of food products, like desserts, candies, and jellies [138]. Gelatin polymer is utilized in the pharmaceutical sector as a wound dressing, covering for capsules and tablets, and drug delivery system [139]. It is a component of hair and skin care products in the cosmetics sector. To enhance the health of the skin and hair, it is a component in many cosmetic products such as face masks and conditioners [140]. It is used to bond silver halide crystals in the photographic process. It is employed in the manufacture of photographic film and paper [141]. Gelatin polymer is utilized in the medical sector as a tissue adhesive, surgical sealant, and hemostatic agent. Diseases of the bones and joints are also treated with gelatin-based materials [142].

2.2. Synthetic biodegradable polymers

2.2.1. Aliphatic polyester

Polyesters are very important materials from rigid to flexible films and hydrophobic to hydrophilic behavior due to their numerous interesting structural properties [143]. Moreover, hydrolysis or alcoholysis (solvolyses) can depolymerize polyesters to their corresponding monomers easily. These chemical recycling techniques are being developed but still have not reached the large–scale level. Stouten and Bernaerts reported the synthesis and different types of properties for numerous aliphatic polyesters (from 1,4-butanediol-succinic acid, lactic acid, or terephthalic acid-based) through condensation polymerization (dehydration) [144]. Aliphatic polyesters can be easily depolymerized by hydrolysis/ alcoholysis compared to their aromatic analogs and are highly employed as single-use goods in packaging, medical articles, and agricultural fields [144].

In addition, Koller and Mukherjee described other types of aliphatic polyesters, poly(hydroxyl alkanoates) (PHA) synthesized in vivo by numerous microorganisms and this can be stored as energy-rich granules when the microorganisms need carbon source similar to starch which is stored in cereal and tubers for later plant/seed growth [145]. Further, Narmon et al. [146] described a ring-opening polymerization (ROP) technique as a scalable and sustainable strategy for the fabrication of high-molecular-weight polymers (*e.g.,* preparation of polyesters from lactones). The functionalization of polyesters is one of the valuable

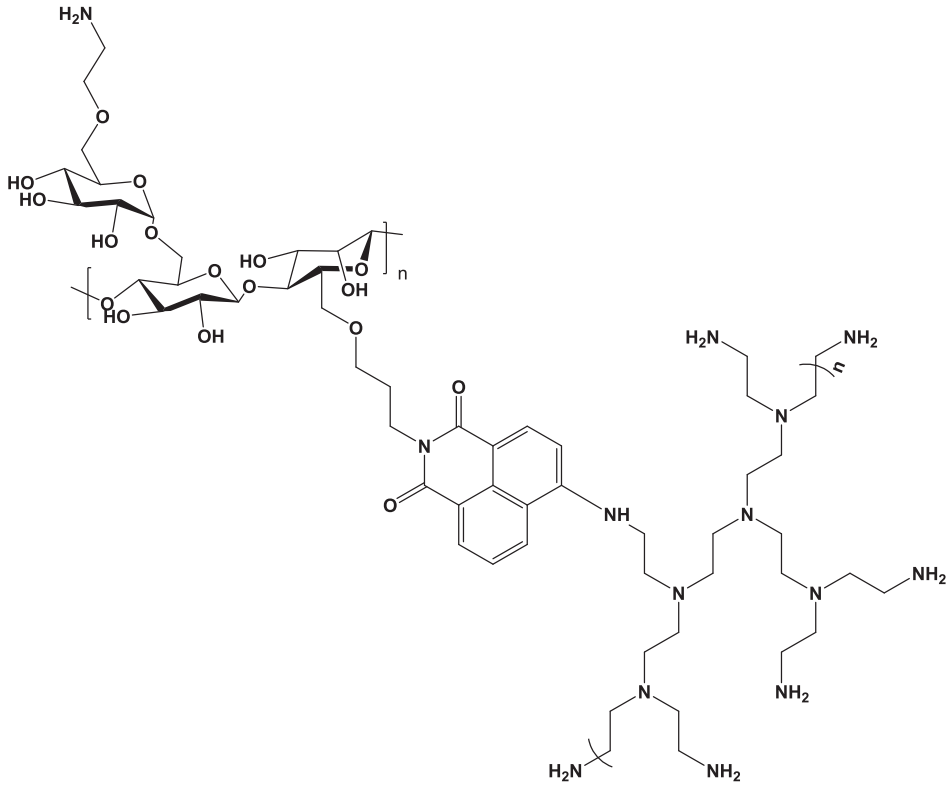


Fig. 24. Fabrication of naphthalic anhydride-coupled guar gum (GG) grafted low molecular polyethyleneimine (LPEI).

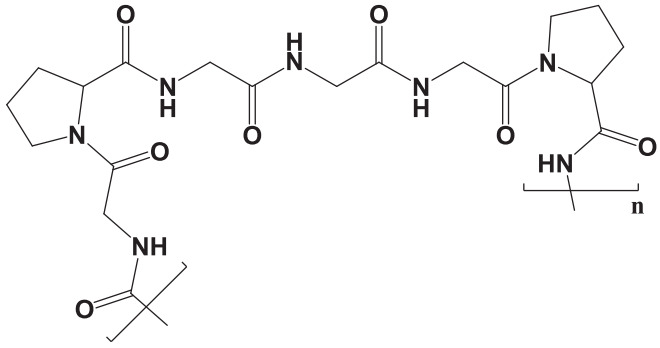


Fig. 25. Structure of collagen.

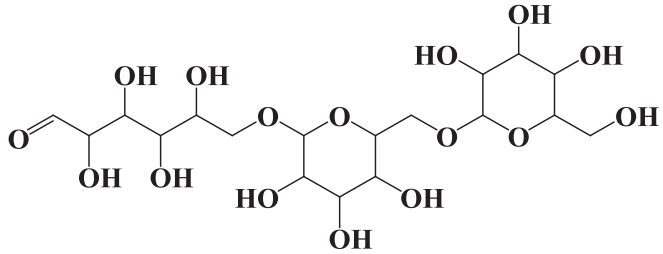


Fig. 27. Structure of dextran.

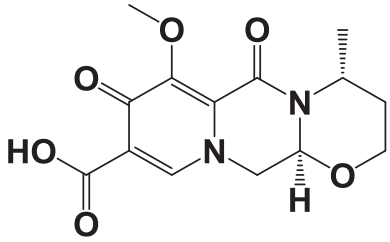


Fig. 26. Structure of albumin.

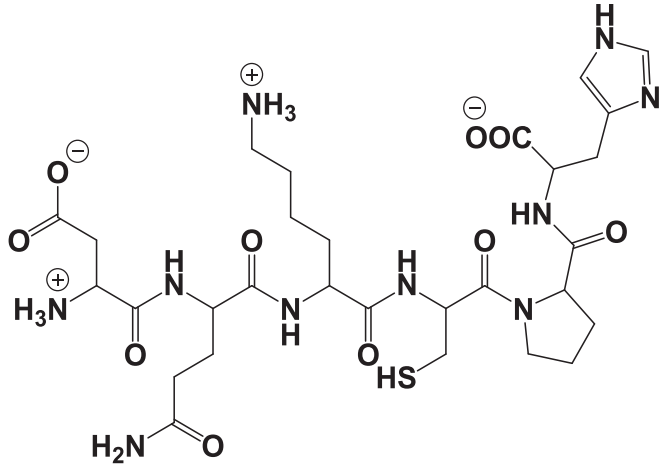


Fig. 28. Structure of Gelatin.

strategies to increase hydrophilicity and lower crystallinity which indicates the biodegradability of a polymer [147].

2.2.1.1. Poly(glycolic acid). As a biodegradable polymer, poly(glycolic acid) (PGA) has the simplest linear structure of an aliphatic polyester with toughness, but the limitation of this polymer is its hydrolytic instability. This polymer has a glass transition temperature (T_g) between

30℃ to 40℃ and a crystal unity around 45℃. This polymer melts (T_m) in the range of 225℃-230℃ and it is also insoluble in water. This polymer can be broken down by certain enzymes present and can also be degraded by hydrolysis (Fig. 29). It is used to deliver drugs (*e.g.,* steroid hormones, antibiotics, and anti-cancer agents) in the form of implants, small microspheres, etc [119,125,131,137,143].

2.2.1.2. Poly(lactic acid). PLA, poly(lactic acid) is an aliphatic polyester and also thermoplastic. It can be derived from various renewable resources like tapioca products (starch, roots, or chips), corn starch, or sugarcane. Under selected conditions, this polymer can be biodegraded easily, like in the presence of lactic acid or CO_2. The results demonstrate that this polymer is highly crystalline with a high melting point, but has low stability. For producing lactic acid by doing bacterial fermentation, corn starch or sugar cane was used. The PLA polymer can be used in the preparation of orthopedic devices or sutures (Fig. 4) [144–147].

2.2.1.3. Poly(caprolactone):. PCL, polycaprolactone polymer, is having a low glass transition temperature (-60° C) and is a really good biodegradable polyester. At around 60° C this polymer starts melting. It degrades very slowly than poly-lactic acid polyester. The duration of its activity is as long as a year for drug delivery. PCL can be used in drug delivery applications including cyclosporin for dental implants in the form of nanoparticles (Fig. 30) [148].

Aliphatic polyesters are the primary basic materials utilized in the manufacture of biodegradable plastics. These plastics are employed in a variety of products, including consumer items, agricultural films, and packaging [149]. They are used to make bone plates, screws, and sutures, among other types of implants. Because they are biocompatible, they do not have any negative effects on the body and gradually degrade and are absorbed by them [150]. Carrier materials for medication delivery systems include aliphatic polyesters. They are perfect for long-term drug administration since they can be made to release medications gradually [151]. Aliphatic polyesters are binders used in the manufacture of adhesives. They have several uses, including labeling, packing, and book binding [152]. They are utilized as coatings in a variety of products, including consumer items, automobile parts, and food packaging. They enhance the tensile strength of coatings, scratch resistance, and adherence [153]. They are used as a raw resource in the textile industry to create fibers. They are utilized in many different contexts, including textiles for industrial purposes, furniture, and clothing [154].

2.2.2. Poly(anhydrides)

The reactivity of this polymer is high and also this polymer is hydrolytically unstable. By using the technique of surface degradation, the polymers can be degraded without any catalyst. The aliphatic polyanhydrides where the $–CH_2$- in the backbone and side chain also can be degraded within days whereas the aromatic polyanhydrides as the side chains can be degraded over several years. The drug-loaded devices for short-term drug delivery can be prepared by using the compression molding or microencapsulation technique. This polymer is currently being used for vaccination and localized tumor therapy (Fig. 31) [148].

Polyanhydrides have been utilized in applications involving tissue engineering like scaffolds for the rebuilding of bone, cartilage, and other tissues. For the development of new tissue, they offer a biodegradable and biocompatible framework [155,156]. The manufacture of implants, like bone plates and pins, uses polyanhydrides. Since they are biodegradable and biocompatible, the body will eventually break them down and absorb them [157]. Polyanhydrides are a key component in the manufacture of adhesives. They can be applied to many different tasks, including bookbinding, packing, and labeling [158]. They are used to make lotions and creams, among other cosmetic items. They provide the products with a velvety smooth feel [159]. Polyanhydrides are a component of pesticide manufacturing. They are perfect for agricultural applications since they can be made to release the poison gradually over time [160].

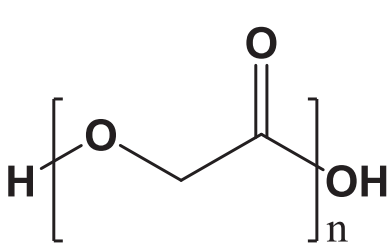


Fig. 29. Structure of Poly(glycolic acid).

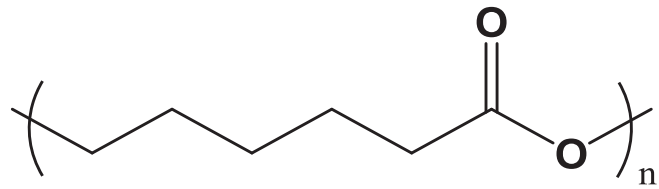


Fig. 30. Structure of poly(caprolactone).

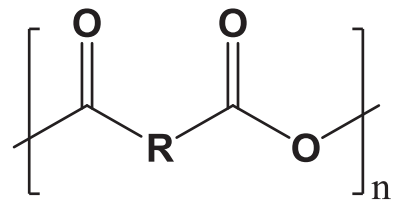


Fig. 31. Structure of Poly(anhydrides).

2.2.3. Poly(phosphazenes):

By using the side groups which are attached to the macromolecular backbone of the polymer, the hydrolytic stability can be determined. This polymer can be used as the material for constructing the soft tissue for tissue-like coatings, and prostheses, and as the material for blood vessels' prostheses. This polymer can be employed in the immobilization of the antigen and the enzymes including drug delivery. Based on the side chain there are 4 types: i) hydrophobic phosphazenes, ii) low Vero cell adhesion, iii) hydrophilic phosphazenes and iv) amphiphilic phosphazenes (Fig. 32) [161].

Polyphosphazenes have been utilized in applications as scaffolds for the restoration of bone and other tissues in tissue engineering [162]. For the development of new tissue, they offer a biocompatible and biodegradable framework [163]. The manufacture of implants, such as screws, and pins, uses polyphosphazene. Since they are biocompatible as well as biodegradable, the body will eventually break them down and absorb them [164]. Polyphosphazenes are utilized as coatings in a variety of applications, including those involving electronic and medical

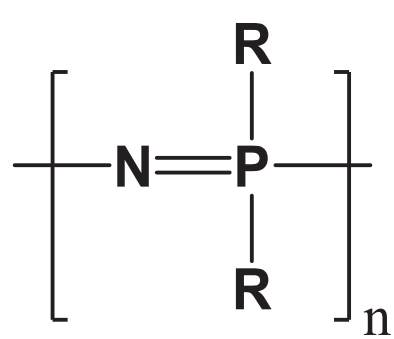


Fig. 32. Structure of Poly(phosphazenes).

devices. They enhance the coatings' biocompatibility, adhesion, and toughness [165]. Polyphosphazenes are employed as flame retardants in a variety of products, including electronics, textiles, and building materials. They increase the materials' fire resistance [166]. Lenses and coatings are all examples of optical materials that are made using polyphosphazene. They are highly transparent and have a high refractive index, which is good optical quality [167].

2.2.4. Poly(amino acids):

Amino acids combine to form a side chain that offers sites where the drug can attach and show its biological activity. This polymer shows a very low level of systemic toxicity because of owning to similarity to naturally occurring amino acids. This polymer can also be used in investigating materials as suture materials. Apart from these important applications, this polymer is also used in producing artificial skins. Apart from the many applications in drug delivery, this polymer has the limitations like restricted solubility, processability, and the unpredictable rate of drug release due to the swelling nature, etc. Those polymers that have more than three or more amino acids have the power to trigger the antigenic response. Polycarbonates which are derived from tyrosine develop as a degradable orthopedic polymer having high strength (Fig. 33) [168].

Polyamino acids are employed in tissue engineering for a variety of purposes, including scaffolds for the restoration of tissues. For the development of new tissue, they offer a biodegradable and biocompatible framework [169,170]. Biodegradable Plastics: Biodegradable plastics can be made using poly(amino acids). They serve as an alternative to conventional plastics made of petroleum, which are not disposable and pollute the environment [171]. They are used to coat a variety of objects, including electronic parts and medical devices. They enhance the coatings' biocompatibility, adhesion, and toughness [172]. Polyamino acids are a key component in the creation of adhesives. They can be applied to many different tasks, including bookbinding, packing, and labeling [173]. They are employed in food as thickeners, emulsifiers, and taste enhancers [174].

2.2.5. Carrageenans

Carrageenans are sulfated linear polysaccharides containing D-galactose and 3,6-anhydro-D-galactose, produced from the extraction of specific red seaweeds of the Rhodophyceae family [175,176].

In general, carrageenans possess 3-linked β-D-galactopyranose units with 4-linked 3,6-anhydro-α-D-galactopyranose or 4-linked α-D-galactopyranose [177]. Based on the degree of sulfation, carrageenans are categorized as κ-carrageenan, λ-carrageenan, and ι-carrageenan. The λ-carrageenan and ι-carrageenan contain two and three sulfates per disaccharide unit, respectively, whereas κ-carrageenan contains one sulfate group per repeating unit. The degree of sulfonation strongly influences the formation of hydrogels *i.e.,* κ-carrageenan exhibits the best hydrogel-forming ability with the least sulfate content among all. (Fig. 34) [178].

Carrageenan is widely used in various products such as dairy goods, processed meats, canned goods, sweets, and other food items as a stabilizing agent to enhance their texture and shelf life [179]. As a thickening agent, carrageenan plays an important role in medicinal goods like cough syrups and digestion aids [180]. Carrageenan has been employed as a gelling ingredient in a range of personal care products, including toothpaste, shampoos and lotions [181]. To increase soil quality and plant growth, it is utilized in agriculture as a biofertilizer and soil conditioner in agricultural activities [182].

Further, several applications of carrageenan have been found in pharmaceutical research associated with medicinal macromolecules for cell migration and tissue repair [183]. Many therapeutic properties have been reported including anti-hyperlipidemic [184], immunomodulatory, anticancer [185], and anticoagulant effects. [186], antiviral qualities preventing hepatitis A virus from reproduction [187] and transmittable human papillomaviruses [188].

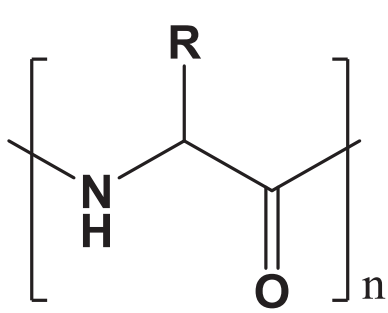


Fig. 33. Structure of Poly(amino acid).

2.2.6. Alginate

Alginate, an anionic biocompatible polysaccharide associated with brown algae, can be made from two copolymers, α-L- guluronic acid and β-D-mannuronic acid [189]. The salient features such as strength, flexibility, high stability, gelling properties, and high viscosity make alginate a remarkable industrial polysaccharide. (Fig. 35).

In general, alginate is used to study integrin signaling in the extracellular matrix (ECM). These biocompatible polymers of alginate have been employed in several medical applications including systems for dressings, bioengineering, drug delivery and implants [190].

Alginate is one of the common gelling as well as thickening agents in the food industry. In addition, it is used to make a variety of food items including dairy, desserts, sauces, and dressings. To improve the stability and bioavailability of food elements like vitamins and minerals, alginate can also be employed in the encapsulation of those nutrients [191]. Further, various pharmaceutical and medical industries use alginate as a drug delivery system by the encapsulation of drugs in alginate microspheres or beads, allowing for regulated drug release. Alginate is also a part of the wound dressing materials to enhance wound healing and prevent infections. Other applications are dental impressions, the creation of tissue engineering scaffolds, and bandages. Also, alginate is a flexible material that can be molded into any shape to create prosthetics [192]. Alginate is extensively used in the biotechnology sector for the immobilization of enzymes and cells. In order to increase the stability and activity of cells or enzymes, it can encapsulate them in beads or microspheres. Further, the production of various bioartificial organs like a synthetic pancreas is one of the most focused research areas [193].

2.2.7. Hyaluronic acid

Hyaluronic acid (HA) is a linear polysaccharide comprising of a repeating disaccharide of (1–3) and (1–4)-linked β-D-glucuronic acid and N-acetyl-β-D-glucosamine unit. (Fig. 36) [194]. Experimental evidence suggests that some HA family members, including HA receptors, HA synthases and HYAL-1 hyaluronidase, support tumor growth. It is an extracellular matrix polymer that occurs naturally in many human tissues. It is nontoxic and biodegradable. It has been widely used for both increasing targeted medication delivery and regulating API release because of its widespread presence in the human body [195]. These molecules are also expressed exclusively in some carcinomas, making them potential targets for cancer cells [196].

In the field of biomedical engineering, hyaluronic acid is utilized to build substrates in tissue engineering and drug delivery systems. It resembles the extracellular matrix found naturally in tissues and organs, hence fostering cell development and tissue regeneration [197]. It is a widely used component in cosmetics, especially in anti-aging treatments. It moisturizes and plumps the skin, hence minimizing the visibility of wrinkles and fine lines [198]. During surgical treatment and other operations, it is employed like a viscoelastic agent in ophthalmology [199]. It offers a desirable alternative for targeted medication administration since it may be altered to target particular tissues or organs. It is used to cure inflammation and discomfort in horses as well

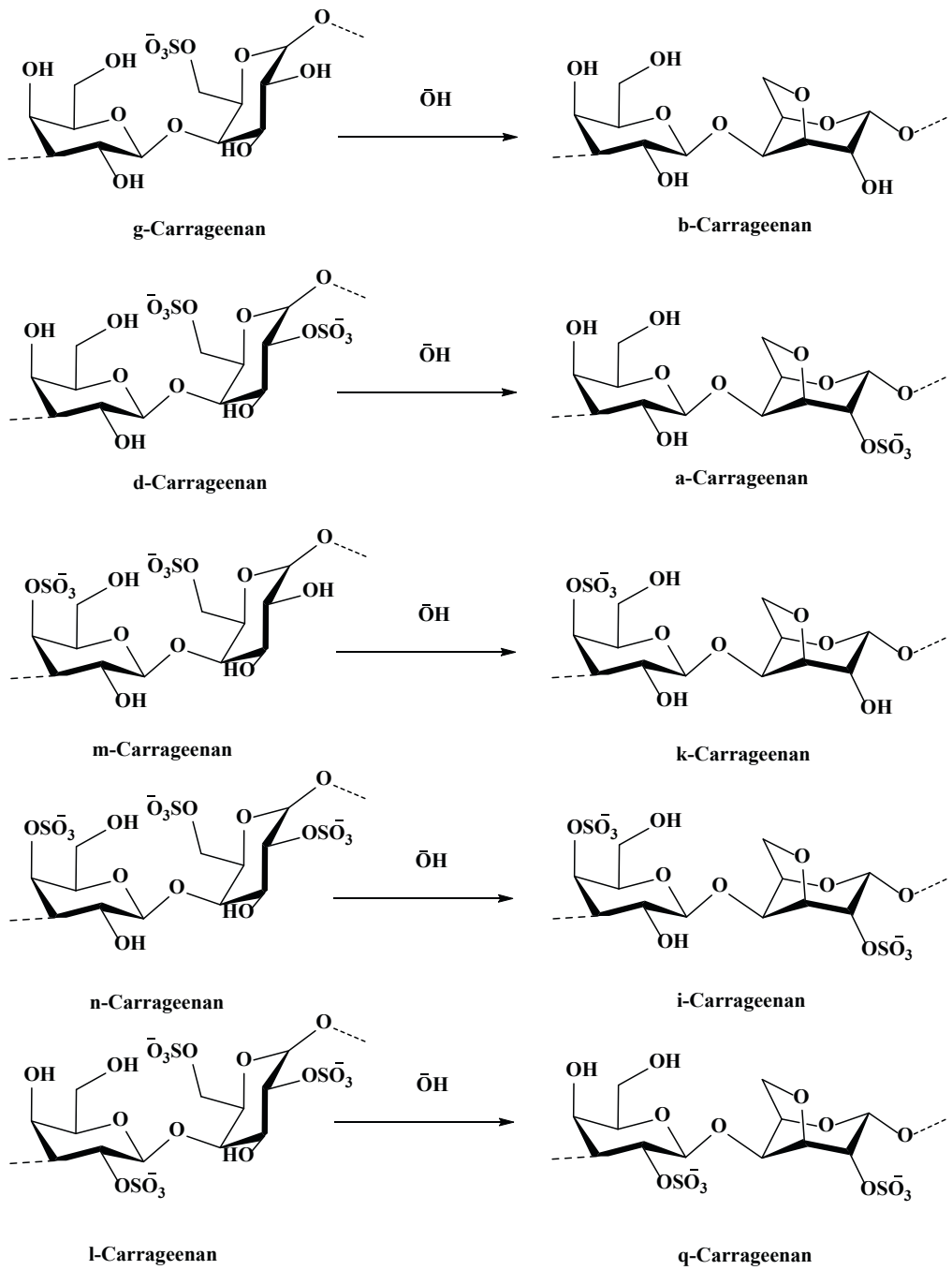


Fig. 34. Conversions from one carrageenan to another carrageenan.

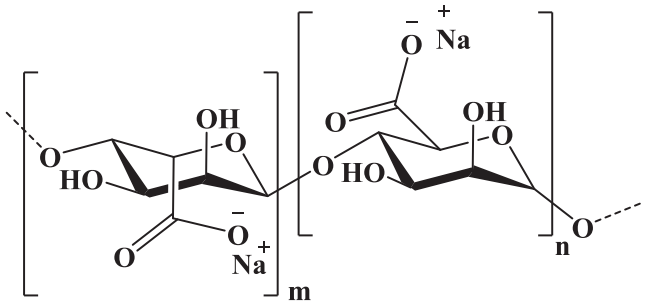


Fig. 35. Alginate.

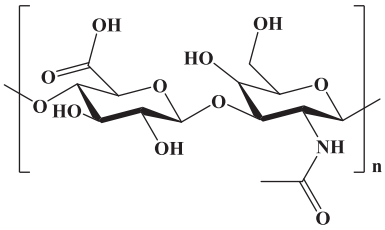


Fig. 36. Hyaluronic acid.

as other animals in veterinary medicine. To lubricate and comfort the joint, it is immediately injected, which lessens discomfort and stiffness [200].

2.2.8. Vitrimers recyclable thermosets

Thermosetting polymers have been widely employed in various elastomers, durable rigid structures, and foams. The biodegradability of thermosetting plastics is one of the most serious issues due to their high stability to the environmental conditions as well as they are difficult to recycle. There is a strong demand for finding suitable methodologies to overcome this problem, but still, the research is in the development stage [201]. In this regard, the current research is focused on the development of covalent adaptive networks (CANs) and, in particular, the vitrimers (associative CAN), which are intrinsically condensation polymers. As demonstrated by Pitet and coworkers, these vitrimers consist of polymers that are cross-linked by reversible covalent bonds and are designed in such a way that there is a reversible open and close cross-linking system and further flexibly upon heating [201]. When these polymers combine with the thermoset materials possessing creep resistance, it is possible to have mechanical recycling by remolding or reshaping. In addition, various types of adaptable cross-links have been developed with heteroatoms in the main chain and with cross-linkers containing remarkable thermal behavior, self-healing and recyclable properties.

Vitrimer polymers, which can heal themselves and attach to a wide range of surfaces, can be utilized as adhesives [202]. Vitrimer polymers can be employed as coatings in a variety of industries, including the automotive and aerospace sectors. They are extremely resilient, abrasion- and corrosion-resistant, and self-healing [203]. Production of plastic packaging such as films, pouches, and containers can be done using vitrimer polymers. Because of their capacity for self-healing and reprocessing, they have the potential to also be reused and recycled numerous times [204]. Textile manufacture is possible with the usage of vitrimer polymers. They are perfect for use in clothes or other textile applications since they can have high mechanical qualities [205]. They can be utilized in the manufacturing of electronic parts like sensors and screens. They may be manufactured without the use of high heat or pressures and have outstanding mechanical and electrical qualities [206]. They can be used to make prostheses and implants, among other medical devices. They are biocompatible and can go through numerous processing steps without losing their tensile qualities [207].

2.2.9. Biodegradable shape memory polymers (BSMPs)

Shape memory materials (SMMs) are a significant class of materials in medicine as they can change shape in response to the ease of medical procedures and new treatment modalities. In addition, shape memory polymers exhibit tunable materials properties, the potential for biodegradation and resorption, excellent shape memory performance, and negligible toxicity. These salient features make these SMMs ultimate candidates for medical field applications because of their remarkable biocompatibility in a variety of biological systems (Fig. 37) [208].

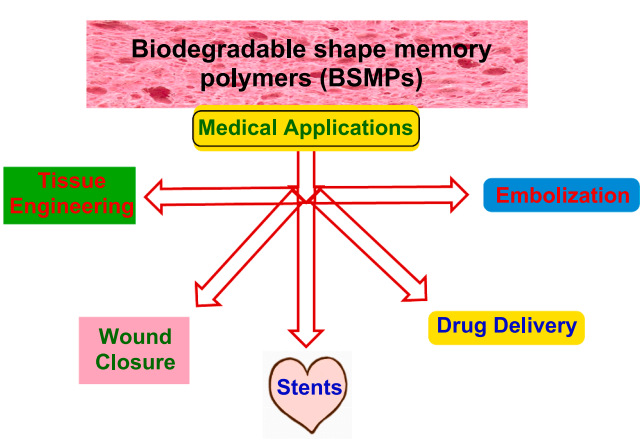


Fig. 37. Remarkable biocompatible applications of BSMPs.

Notably, the nontoxic and biodegradable shape memory polymers (BSMPs) are programmed easily into a temporary shape and can return to a permanent shape upon application of a stimulus [209]. For the implantation applications, in general, the SMPs chosen are easy to process, less dense, lightweight, and economically viable [210].

The two types of cross-links are required for any polymer to have shape memory performance: 1) the stable cross-links forming a permanent network structure through physical/chemical cross-links, and 2) the reversible physical cross-links providing a brief network structure. The stable network forms the permanent shape of an object holding through microcrystalline or glassy domains, hydrogen bonds, and ionic bonds [211]. Various small molecules and polymers useful in the fabrication of BSMPs are given in Chart 1.

In general, the presence of hydrolytically (*e.g.* containing the groups of amides, esters, urea linkages, carbamates, carbonates, etc.) or enzymatically (poly(saccharides) and poly(amino acids)) delicate bonds is responsible for the biodegradation of these polymers. Some of the selected common examples for BSMPs are given in Chart 2.

They are used to make stents and drug delivery systems, among other biomedical devices. They are perfect for specialized drug delivery as well as tissue - engineered applications because they may be created to alter the form in response to particular stimuli [212]. Production of packagings such as films and containers can be done using biodegradable shape memory polymers. They can be designed to alter their shape in response to changes in temperature or other factors, offering a creative and environmentally friendly substitute for conventional packaging materials [213]. Agriculture: Applications for these polymers in agriculture include biodegradable plant supports and controlled-release fertilizers. They can be constructed to alter form according to moisture or other stimuli, acting as a biodegradable replacement for plastic plant supports and a controlled-release device for fertilizers [214]. Textiles: Textiles that can alter their form according to temperature or other stimuli can be made using biodegradable shape memory polymers. This might result in the creation of novel new textile materials for garments and other uses [215]. Environmental remediation applications can use biodegradable shape memory polymers. They can be made to alter their shape in reaction to particular stimuli, which enables them to bind to it and remove specific types of environmental pollutants [216].

3. Preparation methods of biodegradable polymers

3.1. Traditional methods

3.1.1. Modified natural polymers

Poly(saccharide) polymers such as cellulose, starch, chitin and chitosan are known to biodegrade easily and can be converted to other derivatives of biodegradable materials including films by structural modification through blending. The biodegradable films have established improved mechanical properties, low water vapor permeability and solubility.

Depending upon the nature of the source, starch has different amounts of amylopectin and amylose. It is one of the assuring biopolymer materials due to its non-toxicity, biodegradability, renewability, and availability. The starch films can be produced by different molding processes and are very useful in packaging utilization [217]. Structurally modified wood cellulose fibers, such as taurine cellulose, α-hydroxy sulfonic acid cellulose (HSAC), or dicarboxylic acid cellulose, were employed in the production of films to produce environmentally benign materials. These HSAC films exhibit outstanding mechanical properties [218].

Chitosan (CS) is a polycationic polysaccharide known as a linear polymer and non-poisonous material consisting of β-(1 → 4)-linked 2-

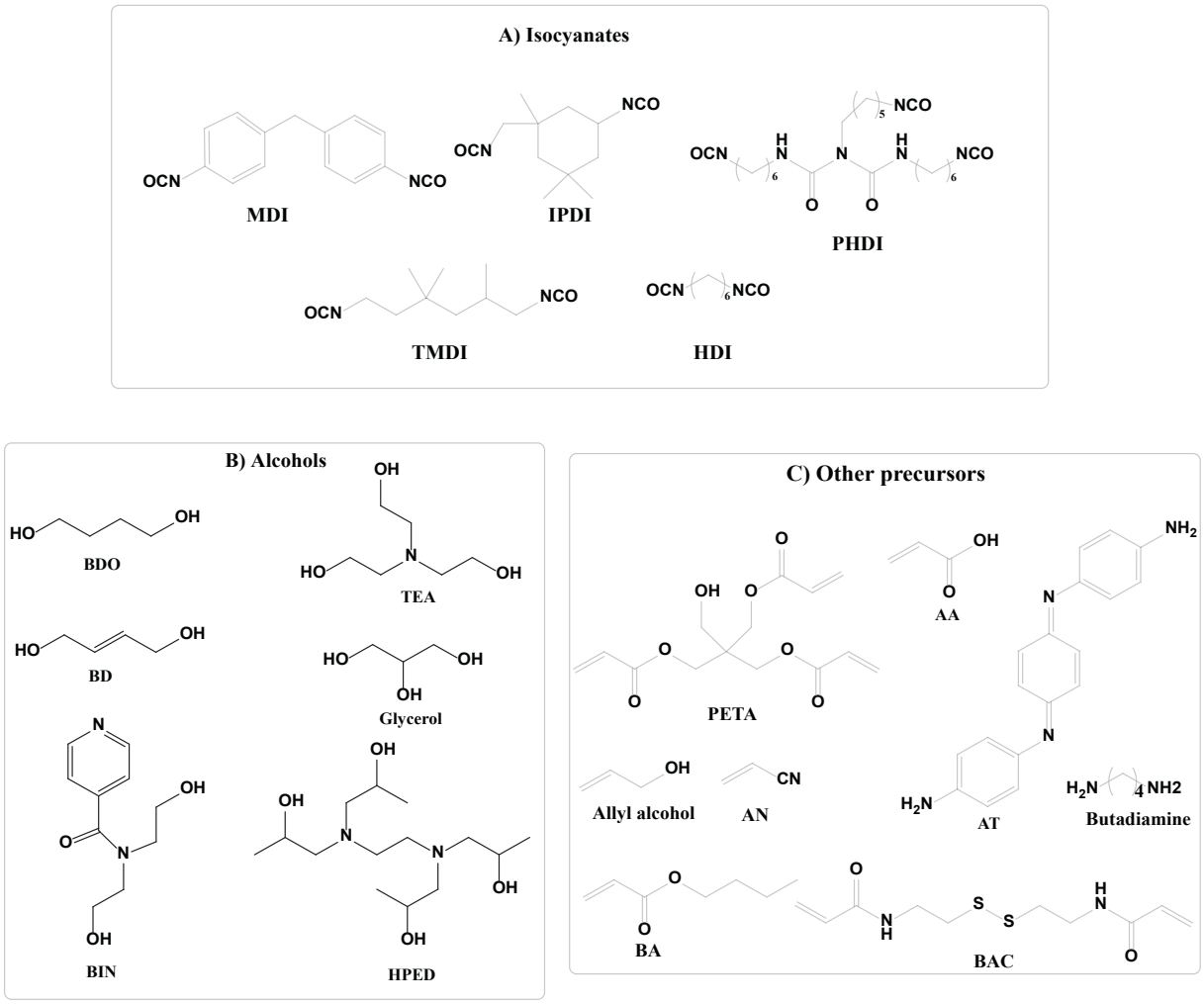


Chart 1. Molecular structures of precursors **A**, **B** and **C** employed to afford the biodegradable SMPs.

amino-2-deoxy-d-glucose (d-glucosamine) and 2-acetamido-2-deoxy-d-glucose (N-acetyl-D-glucosamine) units. This heat-resistant polymer is completely or partially soluble in water and can form films without the use of any other additives. Chitosan has good absorptivity to carbon dioxide and oxygen by displaying outstanding mechanical properties. Moreover, it also has excellent anti-microbial activity and can therefore be used to enhance the shelf life of food [219].

3.1.2. Chemically synthesized biopolymers:

These biopolymers are chemically synthesized as their structures are like natural polymers. These polymeric chains contain easily biodegradable amide, ester, and peptide bonds. Interestingly, a biodegradable material (poly(lactic acid-co-glycerol)) was prepared from L-lactic acid and glycerol as starting materials under specified demanding reaction conditions. This method is economically not viable because of the rigorous reaction conditions as well as the separation of formed many by-products [220].

3.1.3. Microbiologically synthesized polymers:

Microorganisms can synthesize a series of multiplex polymeric materials using some organic matter (*e.g.,* glucose/ starch) as food sources. These polymers contain numerous types of poly(saccharides), poly(esters), silk, etc. due to their similar chemical properties, the separation of the products from the reaction mixture is very complex. Bacterially-synthesized poly(hydroxyl alkanoates) (PHAs) (Fig. 38) as highly biodegradable and biocompatible plastics have enhanced broad applications and can be produced easily many through natural resources [221].

3.2. Enzymatic synthesis:

Enzymatic catalysis is one of the promising techniques for the production of biodegradable polymers. In general, they are very specific to the selected polymers and they catalyze some special polymerization reactions only. Due to the high specificity of the enzymes, the enzymatic polymerization process does not produce any by-products at room temperature and atmospheric pressure and subsequently, the separation of products is easy. Moreover, the enzymes employed can be recycled, and the mild catalytic reaction conditions make the process cost-effective. Some special products based on the stereo specificities of the enzymes can be achieved otherwise they are very difficult to synthesize by other routes. The synthesis of fully biodegradable polymers like aliphatic poly(esters) (by lipase-catalyzed polymerization), poly(ester), poly(saccharide), poly(amide), etc., can also be achieved [222].

3.3. Chemo-enzymatic synthesis:

Since the preparation of high molecular weight biodegradable polymers is very important, chemo-enzymatic synthesis is a remarkable strategy in this regard. Enzymatic synthesis cannot provide the polymer with stereoselectivity and the required high molecular weight cannot be achieved. The chemo-enzymatic synthetic method merges conventional

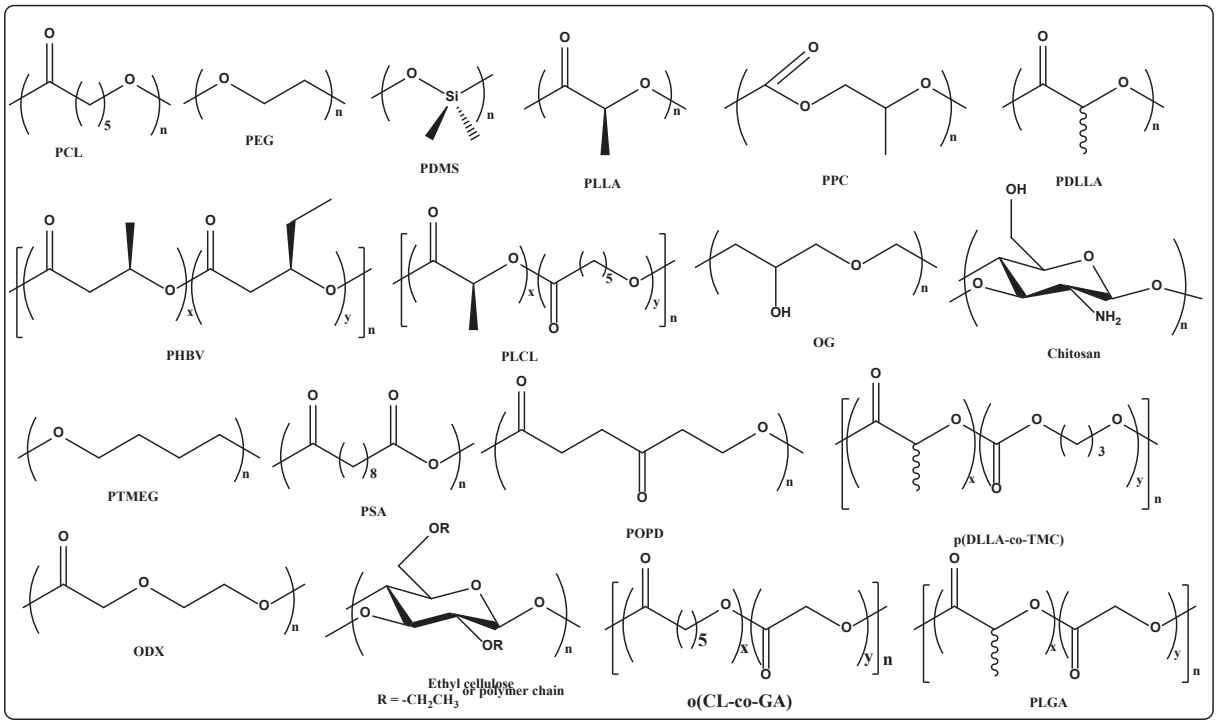


Chart 2. Selected examples of BSMPs exhibiting high biocompatibility.

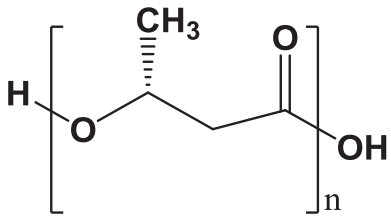


Fig. 38. Structure of Poly(hydroxy alkanoates).

polymerization with a highly regioselective and efficient enzymatic approach [223].

Biodegradation is the chemical degradation of materials by microorganisms such as algae, bacteria, and fungi into environmentally benign products such as biomass including carbon dioxide and water. Biodegradable materials break down into biomass, methane and carbon dioxide. Bacteria that are important in the biodegradation process include Bacillus (capable of producing thick-walled endospores that are resistant to heat, radiation and chemical disinfection), inter alia, Pseudomonas, Klebsiella, Actinomycetes, Nocardia, Streptomyces, Thermoactinomycetes, Micromonospor a, Mycobacterium, Rhodococcus, Flavobacterium, Comamonas, Escherichia, Azotobacter and Alcaligenes [224].

A lot of biopolymers can directly be dumped in soil, composts, or landfills. The materials are broken down only if the desired microorganisms are present. Polymers that are based on normally developed materials such as starch or flax fiber are in danger of corruption by microorganisms. The fabric may or may not decompose most quickly beneath oxygen-consuming conditions by the microorganisms. In the case of materials where starch is included as an additive to a customary plastic network, the polymer in contact with the soil and water is physically attacked by the microorganisms. The organisms digest the starch, and they leave behind a permeable, wipe-like structure with a gigantic interfacial zone, and with a lower structural quality. When the starch constituent has been used, the polymer matrix begins to deteriorate by an enzymatic attack immediately [225]. Some of the polymers undergo degradation by the action of sunlight in the presence of additives like benzophenone. The modification of the structural composition of the polymers can be done in such a way that the UV-absorbing groups can make the polymers biodegradable [226].

4. Properties of biodegradable polymers

One of the biodegradable resins (PVA217, Kuraray Co., Japan) was mainly used in cases where the whole resin is completely soluble in water having the composition of polyvinyl alcohol and the other one is a film of corn-starch which is based on the biodegradable resin (Fig. 39). The usual length and diameter used for biodegradable fibers are 500 and 70 um [227].

4.1. Properties:

4.1.1. Physical properties:

4.1.1.1. Tensile properties:. The biodegradable polymers are cut by using an Autograph AG-1 machine (5kN) at an exact 5 mm per minute crosshead speed. Bamboo fiber is also known as a unique biodegradable

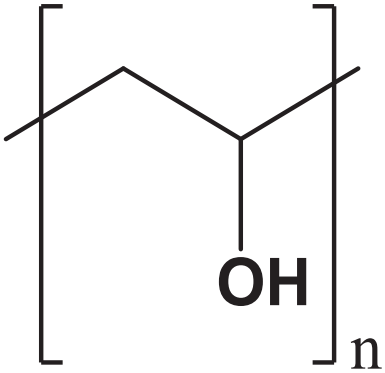


Fig. 39. Structure of Poly(vinyl alcohol).

polymer in textile material industries [228].

4.1.1.2. Water absorption of biodegradable polymer:. For the water absorption test, the sample is dried at 45° C for 24 h and then this sample is used for the water absorption test. Then periodically at 25° C in deionized water, the samples were dipped and taken out. After that, the water which is in excess on the wall of the fiber is blotted with tissue paper and then it is weighed. The equation which calculates the amount of water absorbed (Mt) is as follows,

$$M_t = \frac{W_t - W_o}{W_o} \times 100$$

In this equation, the W_t and W_0 represent the specimen's weight before and after immersion in water respectively [229]. One law called Flick's Second law of diffusion is mostly used in these cases to determine the water absorption test of biodegradable polymers. It can be applied when the material is homogeneous and there is no chemical interaction between water particles and the material.

$$\frac{dc}{dt} = D_{eff} \cdot \frac{d^2c}{dx^2}$$

An effective diffusion coefficient (D_{eff}) is used here to consider the chemical interaction between water particles and the material; where c is the water concentration, D_{eff} is the diffusivity normal to the surface, *i.e.,* in the x-direction, and 't' is the diffusion time [229].

4.1.1.3. Electrical and thermal conduction properties of Bio-degradable polymeric fiber (BDPF):. Biodegradable conductive polymers are new-generation electro-active materials that essentially possess conductive polymers and biodegradable materials. They have excellent optical and electronic properties and are ideal for biomedical applications. Therefore, biodegradable polymers such as poly(caprolactone) (PCL) and poly (D-lactic acid) are linked through ester bonds (Fig. 40). Some approaches have been proposed to overcome the problems of electroactive performances and biodegradability that i) the formation of block copolymers by modified electroactive oligomers of conducting polymers with degradable ester linkages; ii) preparation of polyesters by copolymerization of altered biodegradable and electroactive conducting polymers.

There is a need for the production of biodegradable polycomposite materials with the required electrical and thermal conductivity to reduce the problems associated with composite materials. Using the filler dispersion principle, a mechanically strong, biodegradable and melt processable polymer-based nanocomposite (a combination of graphene nano-platelets (GNPs) poly(butylene adipate-co-butylene terephthalate) was designed and synthesized which displays an excellent electrical and thermal conductivity (Fig. 41) [230,231].

Recently the development of modern technology for producing biodegradable polymers is one of the most current issues. stuffing of the polymer with natural components increased the interfacial contact boundary, through which moisture and aggressive chemicals can penetrate the material, which speeds up the decomposition of the composite in natural conditions. This study is mainly about the study of the thermophysical properties of biodegradable composite materials based on recycled poly(propylene) with different natural fillers of plant origin. The resistivity and thermal conductivity were measured depending upon filler concentration, temperature and residence time in the retrieved soil in which the samples were biodegraded, showing the relationship between the change in thermal conductivity and resistivity of the earlier studied samples and proposed fundamental conclusions about the mechanisms of thermal conductivity of the materials being considered. Adding natural stuffers of plant origin to composites based on retrieved poly(propylene) gradually increased the thermal conductivity coefficient with increasing filler content. In this case, the resistivity gradually decreases, and with an increase in the content of the filler, this decrease becomes more and more significant. This means that the coefficient of thermal conductivity increases with an increase in conductivity, which is typical for materials in which charge transfer occurs at the cost of electrons [232].

4.1.2. Mechanical property

The biodegradable polymers (*e.g.*, poly(glycolic acid), poly(lactic acid), poly(ethylene oxide), poly(dioxanone)) have high mechanical strength and because of their mechanical strength, they can be in many places like internal fixation of broken bones, etc. and their copolymer are having high mechanical strengths. Many studies are done either through in vitro or in vivo processes. These biodegradable fibers provide more strength when compared with stainless steel and they are up to 36% strong compared to that, 3% stiff and 54% bending. Those composites which have carbon or inorganic material in them generally lose strength quickly [233].

Among other biodegradable polymers, poly(lactic acid) plays an important role in textile fabric industries, nonwoven, filler fabric, sanitary napkins, facial masks, baby diapers, etc. because of its good mechanical properties and biodegradable properties. It is synthesized from PLA resin by using melt spinning and electro-static spinning techniques [234]. Strict testing of different metallic polymers and alloys as polymers requires low hardness and loading situations with better control. Those compounds possess a polymer matrix and a reinforcement of less than 20% which can be used for testing under the same conditions [235].

Studies on natural fiber compounds reveal that these can be an effective option for changing glass fiber possessing due to their advantages over conventional mineral fibers including high strength and modulus, improved biodegradability decay, low congestion, increased

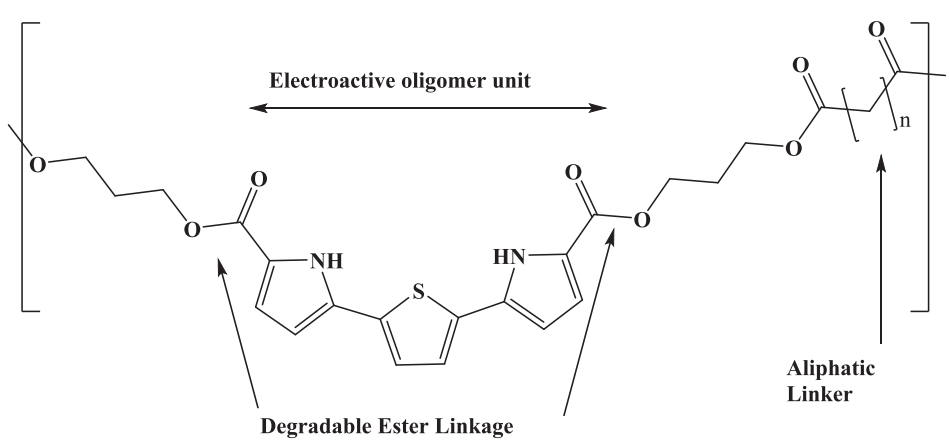


Fig. 40. Structural aspects of a degradable polymer.

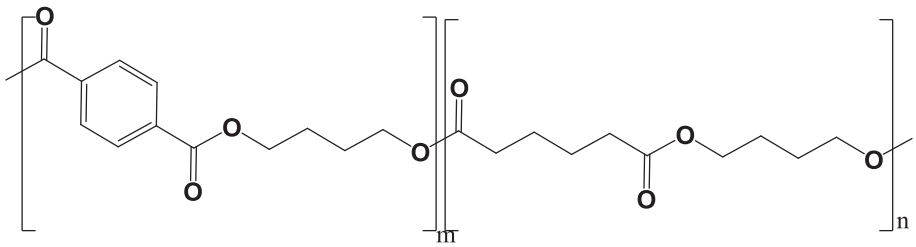


Fig. 41. Structure of poly(butylene adipate-co-terephthalate).

durability, increased noise absorption, environmental purification, and quick regeneration ability. Several polymer composites of naturally reinforced natural fiber have been made and used in sandwich-building programs. For example, Du et al. investigated sandwich panels containing polymer-reinforced paper (PRP) composite as leather material, and natural fiber combined with structure as context. The authors noted that the size of the cell and the height of the honeycomb significantly affected the mechanical properties of the sandwich structure. The results showed that the sandwich structure read here can replace existing commercial products. In recent work, Du et al. produced bio-based sandwich compounds, with leather as well essential elements made of biofiber and poly(lactic acid) (PLA) matrix. The results of the structural analysis indicate that 50 % of the fiber is related to high-pressure structures. A honeycomb templated structure was derived from a combination of reinforced jute vinyl ester through lateral compression molding. In addition, Petron et al. established an eco-friendly honeycomb core by combining fiber with a poly(ethylene) matrix for sandwich panels. Numerous activities and investigations into the mechanical behavior of natural fibers were extensively investigated [236–241].

4.1.3. Optical property

Many techniques for the installation of optical neural implants are based on composite silica or durable polymeric fiber structures. In this regard, non-linear and decaying fibers based on poly(l-lactic acid) (PLLA) were studied (Fig. 42). The PLLA fibers, similar to the size of normal silica fibers, were produced using a simple heat-painting process at approximately 220 °C, which exhibit high mechanical flexibility and visual acuity. The appearance of structural and material changes was thoroughly examined during in vitro degradation. Furthermore, their biocompatibility and brain tissue were tested in live mice, and full degradation was demonstrated to use them as a tool for intracranial light delivery and detection of deep brain fluorescence sensations and optogenetic investigations in vivo. The introduced materials and the device platform provide access to fully compatible bioresorbable photonic systems for biomedical use [242,243].

5. Factors influencing the properties of biodegradation of polymers

The physical and chemical factors influence the biodegradation mechanism of the polymers based on their significant surface area. Various factors influencing the rate of the biodegradation process of polymers are based on the features of polymers and the exposure conditions. The properties of polymers like crystallinity, nature of the functional groups, surface morphology, degree of cross-linking, nature of different monomers (co-polymeric structures), flexibility, blending, tacticity, molecular weight and additives employed. The possible exposure conditions for abiotic are pH, UV radiation, moisture and temperature and the biotic conditions are mostly hydrophobicity, the presence of biosurfactants and extracellular enzymes. The degree of biodegradability strongly depends on the availability of hydrophilic end-groups and the high surface area of the polymers [244,245].

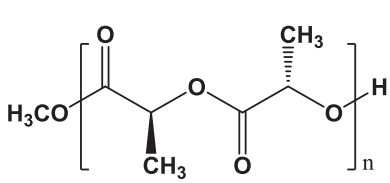


Fig. 42. Structure of Poly(L-lactic acid).

In polyester, the side chains degrade more slowly than the main chains. Side chains are not included because molecular weight is what decides many of the polymer's physical properties, so plays a significant impact in deciding its success in biodegradability. If the molecular weight of the polymer is increased, then the biodegradable property of this polymer decreases [246]. Notably, the morphology of the rates of a polymer has a significant and crucial impact on biodegradation. This is due to the enzymes attacking the amorphous parts of a polymer, other than these things a very significant component in biodegradability is the degree of crystallinity because the amorphous parts in the polymer are loosely packed therefore it is more susceptible to deterioration nonetheless. The crystalline section of the polymer is relatively more resistant than the non-crystalline part due to the denser packing of the amorphous area molecules. So, if the crystallinity increases in the polymer then the degradation of PLA decreases over time. There are a few factors on which the enzymatic degradation of the polymers depends and the melting temperature (T_m) of the polymer is one of them.

$$T_m = \frac{\Delta H}{\Delta S}$$

ΔH = Enthalpy change on melting and ΔS = Entropy changes in melting.

There are some general methods by which the evolution of biodegradability is determined. Increases in characteristics such as macro molecule molecular weights, hydrophobicity and crystallinity, for example, reduce biodegradability. Now water repellant property is called hydrophobicity and getting attracted by water is called hydrophilicity. So, when a polymer is having both these properties, then the polymer exhibits good biodegradability. Synthetic polymers having long repeating unit coating tend to crystalize and biodegrade less likely; still, a polymer Subtilisin was found to have the property to quickly digest a series of poly(amide-urethane). Other factors include [247] processing conditions, shape, the presence of unexpected units or chain defects, distribution of molecular weight, physicochemical factors (ionic strength, ion exchange, and pH), annealing, adsorbed and absorbed compounds (water, ions, lipids, etc.) storage history, sterilization process, the configuration structures, site of implementation, and mechanism of hydrolysis.

Notably, PMMA (polymethyl methacrylate) is a biocompatible synthetic thermoplastic polymer and is widely used in various applications such as construction, automotive, and medical devices. However, it is generally not considered as a biodegradable polymer because it does not readily break down into natural components under typical environmental conditions. As it was found in recent reports, the structural changes in PMMA facilitate biodegradability [248].

6. Applications of biodegradable polymers

Plastics with similar longevity properties are ideal for different types of applications, including building materials, commodities, packaging, and as well as hygiene products. This can cause serious waste disposal issues in the future because, these traditional petroleum-derived polymers are non-biodegradable, and they are resistant to microbial degradation. Biodegradable polymers denote a developing field toward a cleaner environment as they are derived from renewable feedstocks, and reduce greenhouse gas emissions.

A large number of biopolymers have been employed as one of the major constituents in medicine, the automotive, agriculture, and packaging industries. As these kinds of materials receive more consideration and support from all levels of government to decrease the volume of inert materials now being thrown into landfills, more biodegradable polymers with different blends and required properties will be developed. The three underlying segments for the utilization of biodegradable polymers are medicine, agriculture, and bundling. Due to having decomposability, biodegradable polymers cannot be used directly in pharmacological devices, like lattices for chemical immobilization and controlled-release gadgets but too recuperative devices, such as brief prostheses, and permeable forms in tissue engineering.

6.1. Applications in medicine and pharmacy

The current implementation of biodegradable polymers includes orthopedic surgery and vascular applications. Because of the achievable required toughness and a movable extortion speed, biodegradable polyesters are broadly used as implantable lattices inside the body for the controlled release of drugs and also promising porous structures in tissue engineering. Another natural polymer, gelatin is currently being used for coatings and micro-enclosing different drugs for biomedical applications including the preparation of biodegradable hydrogels. The PHBV (poly(3-hydroxybutyrate-co-3-hydroxy valerate)) material exhibits a distinctive piezoelectric property and is utilized in applications where electrical re-enactment is linked.

Synthetic polymers are broadly employed in various gadgets and biomedical implants since they can be assembled into different forms. The non-woven PGA textures have been scrutinized as frameworks for tissue rejuvenation. The use of chitin and its subsidiaries are involved in anti-cholesterolemic agents, drug carriers, anti-tumor objects and blood anticoagulants. Collagen and chitin-based poly(L-leucine) are remarkable sources for skin surrogates or wound dressing materials. Since the cell-sticking property is shown by PLGA, it can be used for applications in tissue engineering. Poly(anhydrides) and PLGA materials are used in controlled drug delivery systems [249–251].

6.2. Application in packing:

The usage of biodegradable polymers as package materials is very well-known for a long time due to their remarkable properties such as air permeability and low-temperature scalability. The PLA has a medium permeable level to water vapor and oxygen and is utilized in bundling implementations such as bottles, cups, and films. Chitosan is used extensively in paper-based packaging such as a coating for fat barriers, to produce oil blockade packaging, etc. Films based on chitosan have proved to be very effective in food storage due to their anti-microbial properties [252–255].

The PHBV materials (poly 3-hydroxybutyrate-co-3-hydroxy valerate) were prepared by using ceiba pentandra fibers. Because of its thermos-mechanical property, hydrophilicity, transparency, anti-microbial property and biodegradable property, it can be used as a packaging material [256]. By using electrospinning, the appropriate anti-microbial fibers can be made using non-toxic organic solvents from zein. This biodegradable polymer can be used in food packaging. It secures the food which is inside of it without decreasing its quality of it [257].

6.3. Application in agriculture:

Biodegradable materials are promising candidates for the development of intensified sustainability, mulching, and low-tunnel soil due to their susceptible properties. Biopolymers are largely used in making ropes and angling nets in marine horticulture and other environmentally friendly agricultural activities. When starch is kept in contact with it, it gets degraded. This is the reason why starch films are usually used as agricultural mulch films as they produce non-toxic products while interacting with soil microorganisms [258].

6.4. Application in the geotextile industry:

Geo-textile fibers are really useful in today's world as they are made to determine the soil stability of a place. Biodegradable fibers can be used as a substitute for synthetic geotextile fibers because of their high mechanical property and low cost. Also, biodegradable polymers degrade into various components which don't affect mother nature. It has been illustrated that these polymers can act as useful fertilizers for plants after degrading into fertilizer components. The life span of a geotextile fiber is a couple of months to a few years. It depends on the environment like heavy rains, microbiological attacks, etc. the activity of the microbes decreases by increasing the lignin content present in the fiber. Lignin is the main component of a plant's cell wall [259].

7. Processing of natural & synthetic polymers into different forms

Various forms of natural and synthetic biopolymers such as membranes, fibers, hydrogel, scaffolds, beads, and nanoparticles have been extensively employed for their applications.

7.1. Membrane:

Both synthetic and natural polymers can be treated in a variety of shapes, including membranes. There are many uses for polymer-based membranes, including medicine delivery, gas separation, and water filtration. Techniques including phase inversion and casting are frequently used for converting synthetic and natural polymers into membranes. The polymer is dissolved in a solvent during phase inversion, and it is then precipitated by altering the solvent's content or temperature. As a result, a porosity membrane with regulated pore size and shape is created. A method called electrospinning utilizes an electric force to spin tiny fibers from a suitable solvent. These threads can be gathered on a surface to create a porous membrane with a large surface area. In casting, a polymer solution is poured onto a foundation and allowed to dry into a flat membrane. Using this method, membranes with excellent selectivity and precise pore size can be created. Membranes for biomedical purposes are frequently made from natural polymers like cellulose, chitin, and chitosan, whereas synthetic polymers like polyethylene, and polypropylene are utilized in industrial processes like gas separation and water purification [260].

7.2. Fibers:

Fibers made from both synthetic and natural polymers can be used for a variety of purposes. Melt, dry and wet spinning, and electrospinning are some of the processing techniques. To create continuous fibers, melt spinning involves dissolving the material and extruding it via a spinneret. For synthetic polymers like nylon, polyester, and polypropylene, this technique is frequently employed. In dry spinning, the polymer is dissolved in a solvent, and the solution is then extruded via a spinneret and into a hot air chamber. The polymer hardens into fibers as the solvent disappears. This technique is frequently applied to synthetic

polymers like rayon and acrylic.

In wet spinning, the polymer is dissolved in a solvent before being extruded into the coagulating bath, in which the solvent is swapped out for a non-solvent. As soon as the polymer comes into touch with the non-solvent, it solidifies into fibers. This technique is frequently applied to organic polymers like proteins and cellulose. A polymer solution mixture is drawn into extremely tiny fibers during electrospinning and then placed on a substrate. This technique may create fibers with sizes ranging from nanometers to micrometers and is frequently utilized for both synthetic and natural polymers. The polymer that is utilized, the processing procedure, and the processing circumstances all affect the qualities of the fibers that are produced. The processing settings can be changed to control the diameter, strength, and flexibility of the fiber. While synthetic polymer fibers, like polyester and nylon, are widely utilized in a variety of applications, involving clothing, and industrial materials, natural polymer fibers, like cotton and wool, are frequently employed in textiles [261].

7.3. Hydrogels:

Hydrogels consist of three-dimensional systems of polymer chains made from polymers that are both synthetic and natural that can soak and hold a lot of water. Applications for hydrogels include medication delivery and wound healing. Natural and manmade polymers are commonly processed into hydrogels using methods like chemical as well as physical crosslinking and self-assembly. In order to create the hydrogel network, physical crosslinking uses intermolecular forces like hydrogen bonds, van der Waals force, or electrostatic interactions. Natural polymers like agarose and alginate are frequently made using this technique. Chemical crosslinking entails utilizing a crosslinking agent to covalently bond the polymer chains when working with synthetic polymers like polyethylene glycol (PEG).

Using light to start the linking of the polymer matrix is known as photopolymerization. For synthetic polymers like PEG and methacrylate-based polymers, this technique is frequently employed. The polymer chains are put together into a hydrogel network through self-assembly using non-covalent interactions like electrostatic or hydrogen bonding. This technique is frequently applied to natural polymers like hyaluronic acid and chitosan. The processing method, the processing circumstances, and the polymer that was employed all affect the final hydrogels' qualities. By changing the processing conditions, hydrogel qualities including swelling ratio, mechanical properties, and biodegradability can be managed. While hydrogels manufactured from synthetic polymers are frequently utilized in commercial applications like moisture absorption in diapers, hydrogels derived from natural polymers are typically friendly and can be employed in biomedical applications [262].

7.4. Scaffolds:

A Scaffold (*i.e.*, a model for tissue formation) can be created from both natural and manufactured polymers. The extracellular matrix of organs and tissues can be mimicked by scaffolds, allowing for tissue regeneration and cell infiltration. Scaffolds made from natural and synthetic polymers are frequently created using methods including electrospinning and porogen leaching. In order to create fine fibers gathered on a material to create a scaffold, electrospinning is used. 3D printing involves using a computer-controlled printer to create a scaffold layer by layer using a polymer ink or a mixture of polymers and cells. This method allows for precise control of the scaffold geometry and mechanical properties. Porogen leaching is the process of introducing pores into a polymer scaffold by utilizing a sacrificial substance, like salt or sugar. A porous structure with interconnecting pores is left behind after the porogen material dissolves. The polymer that is utilized, the processing procedure, and the processing circumstances all affect the qualities of the scaffolds that are produced. The processing settings can be changed to influence the qualities of the scaffold, like porosity, mechanical characteristics, and biocompatibility. Collagen and chitosan scaffolds are frequently employed in applications involving tissue engineering, whereas PLGA and PEG scaffolds are used in a variety of applications [263].

7.5. Beads:

Beads, which are little spherical or irregular in shape particles with a variety of dimensions and qualities, can be created from both natural and manmade polymers. Many applications, such as chromatography, medication delivery, and biomedical implants, can make use of beads. Emulsion polymerization and suspension polymerization are common methods used to transform natural and synthetic polymers into beads. Emulsion A monomer is dispersed in an aqueous solution including a surfactant to create an emulsion during polymerization. The surfactant then starts the polymerization process, creating beads of polymer that are stabilized. Common synthetic polymers that use this technique include styrene and methyl methacrylate. A monomer is dispersed in water in the presence of a stabilizing agent to create a suspension during suspension polymerization. The stabilizing ingredient subsequently starts the polymerization process, resulting in the formation of polymer beads. For synthetic polymers like polyvinyl chloride and polyethylene, this technique is frequently employed. The polymer is dissolved in an organic volatile solvent, and the polymer solution is then dispersed in an aqueous solution containing a surfactant to create droplets. Following that, the solvent is expelled, resulting the formation of beads. This technique is frequently applied to natural polymers like chitosan and alginate [264].

7.6. Nanoparticles:

Nanoparticles can be created from both natural and artificial polymers. Many applications, such as drug delivery and biosensing, can make use of nanoparticles. Nanoprecipitation is a frequently used process in the creation of nanoparticles from natural and manmade polymers. A polymer is dissolved in a solvent, and then the polymer solution is quickly added to a non-solvent to cause the polymer to precipitate as nanoparticles. This technique is frequently applied to natural polymers like gelatin and chitosan. A polymer and a medicine are dissolved in an organic solvent that is immiscible with water, and the mixture is then emulsified with an aqueous phase that contains a surfactant to create droplets. The drug-containing solid polymer nanoparticles are then left behind once the organic solvent has evaporated. Certain synthetic polymers, such as poly(lactic-co-glycolic acid), are frequently made using this technique (PLGA).

Electrostatic assembly is the process of creating nanoparticles from oppositely charged polymers via electrostatic interactions. By combining liquids of copolymer with opposing charges, this technique causes the creation of particles by modifying the pH of the mixture. This technique is frequently applied to synthetic polymers like chitosan and poly(ethylene glycol). The polymer that was utilized, the technique of processing, and the process conditions all affect the characteristics of the finished nanoparticles. By changing the processing parameters, it is possible to change the attributes of nanoparticles, including their size, shape, and surface charge. Nanoparticles created from synthetic polymers like PLGA and PEG are employed in a variety of applications. In contrast, nanoparticles derived from natural polymers like chitosan and gelatin are frequently used for drug delivery applications [265]. Table 1 highlights a concise description of the applications and recent developments of various biodegradable polymers.

8. Safety and efficacy

One of the most challenging aspects is the usage of biopolymers as biodegradable-polymer drug-eluting stents (BP-DES). Notably, the

Table 1
A summary of applications and recent developments of various biodegradable polymers.

Type of the biopolymer	Applications	Recent developments	[Ref]
Poly(saccharides)	Traditional textile films/ fibers	Fabrication of polyolefin-like plastics, adhesives, unique optical transparency, or high-performance engineering plastics	[63–67]
Lignin	A partial replacement of phenol in Bakelite wood-based glue	Radical-generating enzymes (laccases found in white-rot fungi)	[68–77]
Chitosan	Biomedical field as non-absorbable sutures, drug delivery, vaccine delivery, anti-bacterial agent	Molecular engineering, tissue adhesion and wound hemostasis and dressing and also for bone disorders	[78–90]
Starch	Stabilizers and thickeners are the basic usages of food starch and confectioneries	Drug delivery, oil recovery through chemical flooding technologies, tissue engineering, and diagnostic imaging	[91–100]
Cellulose	Cosmetic and pharmaceutical industries, fibers and clothes	Drug delivery, wound dressing and tissue engineering, antimicrobial application, enzyme capsulation, nucleic acid extraction.	[101–110]
Gaur gum	Many health issues like constipation, diarrhea, high cholesterol, and diabetes.	Non-viral gene delivery to triple-negative breast cancer, drug delivery carriers for protein base drugs.	[111–118]
Collagen	Sponges for burns/wounds, ophthalmology	Majorly used in molecular drug delivery system	[119–124]
Albumin	Drug release, degradation of albumin	High local drug concentration for a relatively long time in chemotherapy	[125–130]
Dextran	Used as a packaging material and polymer coating material because of its biodegradable activity	Used as drugs, especially as blood plasma volume expanders	[131–136]
Gelatin	Tissue engineering, capsule material, cosmetics, ointment	Coating materials and oral controlled delivery of the drug	[137–142]
Poly(glycolic acid)	Absorbable screws and staples, meshes, and stents	Implants and drug delivery in the form of small microspheres	[119,125,131,137,143]
Poly(lactic acid)	Used as industrial and household wipes, diapers and disposable garments, and feminine products.	These polymers are useful in the preparation of orthopedic devices or sutures	[144–147]
Poly(caprolactone)	It is usually used for resins so that it can improve the processing properties like toughness, flexibility, and tear strength.	Nano-drug delivery applications of PCL including cyclosporin	[148–154]
Poly(anhydrides)	Because of its good tissue biocompatibility, it can be used in drug delivery systems.	Drug-loaded devices for short-term drug delivery, localized tumor therapy, and vaccination	[155–160]
Poly(phosphazenes)	Used mainly as coating materials on the surface.	Useful for drug delivery investigations	[161–167]
Poly(amino acid)	Because of its biodegradable activity, it can be used in bone regeneration therapy.	Promising materials for producing suture and artificial skins	[168–174]
Carrageenans	Used as a stabilizing agent to enhance their texture and shelf life	Anti-hyperlipidemic, immunomodulatory, anticancer, anticoagulant effects	[179–188]
Alginate	Used as thickening agents in food industry	Dental impressions, creation of tissue engineering scaffolds, and bandages	[189–193]
Hyaluronic acid	Utilized to build substrates in tissue engineering and drug delivery systems	This can be used as a viscoelastic agent in ophthalmology, component in cosmetics, especially in anti-aging treatments	[195–200]
Vitrimers	–	It can heal wounds or scratches easily and rubbers that are made by synthesis can be welded by this	[201–207]
Biodegradable shape memory polymers	–	These polymers can be easily programmed into a momentary shape and can return to a permanent shape upon application of a stimulus	[208–216]

advancement of these biopolymers has been witnessed in the recent past as powerful due to their safe and long-lasting-polymer drug-eluting stents (DP-DES). Methods for the system of biodegradable microparticles for shipping protein tablets, which include botulinum toxins, were developed. The strategies consist of the stairs of precipitating and washing proteins with natural solvent to dispose of water previous to dispersing in polymer-dissolved natural solvent to save you publicity to water/solvent interfaces and hold bioactivity of the protein tablets and fabrication of microparticles through both template or emulsion method. Biodegradable microparticles, shaped of 1 or extra biodegradable polymers having entrapped inside the polymer one or extra protein agents, which include botulinum toxin, also are provided [269].

Toxin-loaded microparticles can also be formulated into thermals or crosslinked hydrogels. The stability of the protein within these microparticles, as well as the controlled release of the entrapped agents, provides for sustained efficacy of the agents. First-era drug-eluting stents have raised worries concerning the danger of past due and past due stent thrombosis as compared with naked steel stents and require extended twin antiplatelet therapy. Despite sizeable investigations, the physiopathology of those past-due activities stays incompletely understood. Aside from patient- and lesion-associated danger factors, stent polymer has been mentioned as one of the ability causes. The staying power of long-lasting polymer after the whole drug launch has been proven to be chargeable for hypersensitive reactions and inflammatory reactions. Third-era drug-eluting stents with greater biocompatible or biodegradable polymers have finally evolved to cope with this problem [269].

The strength and durability of biodegradable materials may be restricted, which may reduce their value in some important medical applications [266]. Inconsistent rates of biodegradation can be challenging to anticipate and manage the degradation of biodegradable materials in the body since the pace of degradation can vary greatly depending on the particular material and its use [267].

Toxic byproducts released by some biodegradable materials during the degradation process can be detrimental to the body and prevent their use in some applications [266,268]. The use of biodegradable materials in some applications may be constrained by the fact that they may not be suitable with implants and other existing medical equipment. It will take continued study and development to increase the strength, uniformity, durability, and suitability of biodegradable materials in addition to initiatives to speed up regulatory approval procedures and promote their usage in the medical industry to address these issues.

9. Conclusion

The research on exploring biodegradable polymers has been one of the most intriguing areas because of their important association with environmental safety (as packaging materials for the food industry) and their remarkable applications in the medical and agricultural fields including wound healing, drug delivery, etc. Also, these fine-tined biodegradable polymers aim to find a replacement for toxic and unwanted synthetic single-use plastics. Biopolymers or renewable resource-based biopolymers include carboxymethyl cellulose, pectins,

hemicellulose, starch, pullulan, xanthan gum, agar, guar gum, alginate, gum karaya, and gellan, etc. Several copolymerization methods (*e.g.,* block, random or grafting) have been proposed and employed to improve the properties of biodegradable polymers. Another frequently used method is physical blending to produce biodegradable polymeric products with different physical and morphological properties. Notably, these methods produce biodegradable polymers that possess both an enhanced rate of biodegradation and the required mechanical strength of the product derived. There is a strong and increased demand for the sustainable development of new biopolymers whose properties are in close agreement with traditional polymers for industrial applications. One of the major draw-backs of this development is expensive industrial processes. The research on PLA silk and chitosan-based biopolymers has been increasing due to their biocompatibility and suitability for implantable applications in the medical field. Still, the usage of biodegradable polymers is less than 1% compared to synthetic plastics.

10. Future prospects

The development of new biodegradable polymers needs to be simple, cost-effective, and recyclable (if possible) and this also provides huge opportunities in terms of employment. Also, a lot of developments of new hybrid systems are expected soon as these new biodegradable polymers may possess effective and important properties.

The development of structurally fine-tuned BSMPs including identifying new polymer classes, with the appropriate characterization of properties of materials and their performance in vivo is one of the most anticipated areas of research. Though many biodegradable polymers like polyacetals, polyphosphazene, polyesters and polyanhydrides are known, important properties like shape memory are not demonstrated. This is one of the research areas as the search will go on up to the discovery of the fine-tuned and ideal biopolymers to achieve the temporal control of shape, required mechanical properties, biodegradable rates, and other important biocompatibility to the tissues and blood and non-inflammation.

Most of the established studies reported with BSMPs are at the early stage of development and further In vivo studies are required to support the validity of the new treatment modalities, drug delivery applications, drug-eluting, and retractable stents.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

We are thankful to DAE-BRNS (INDIA) (No. 58/14/19/2021-BRNS/37236 and VIT-SEEDGRANT for the encouragement and financial support.

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